Qualitative Dependent Variables

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Contents

2 Qualitative Dependent Variables **3**

2.1 Introduction . . . 3

2.2 Binary outcomes . . . 3

2.2.1 The linear probability model . . . 3

2.2.2 A latent variable model for binary variables . . . 5

2.2.3 ML estimation . . . 10

2.2.4 Interpretation . . . 11

2.2.5 Goodness of fit . . . 17

2.2.6 Example: Labour force participation (Mroz's study) . . . 20

2.3 Ordinal Outcomes: Ordered Logit and Ordered Probit . . . 40

2.3.1 A latent variable model for ordinal variables . . . 40

2.3.2 ML estimation . . . 43

2.3.3 Interpretation . . . 43

2.3.4 Example: Ordered logit - working mothers and childern . . . 48

2.4 Nominal Outcomes: Multinomial Logit . . . 59

2.4.1 The MNLM as a probability model . . . 60

2.4.2 The MNLM as a Discrete Choice Model . . . 61

2.4.3 Example: Nominal outcome - occupation and education . . . 62

2.4.4 ML estimation and testing . . . 65

2.4.5 Example: Nominal outcome - occupation, education, experience and race . . . 66

2.4.6 Interpretation . . . 70

2.4.7 The conditional logit model . . . 78

2.4.8 Example: Conditional logit - transportation mode . . . 81

2 Qualitative Dependent Variables

2.1 Introduction

This chapter considers regression models that are appropriate when the dependent variable is binary, ordinal or nominal.

- **Binary variables** have two categories and are often used to indicate that an event has occurred or that some characteristics is present. Is an adults a member of the labor force? Did a citizen vote in the last election?
- **Ordinal variables** have categories that can be ranked. Surveys often ask respondents to indicate their agreement to a statement using choices strongly agree, agree, disagree, and strongly disagree. Items asking the frequency of occurrence might use the categories often, occasionally, seldom, and never.
- **Nominal variables** occur when there are multiple outcomes that cannot be ordered. Occupations can be grouped as manual, trade, blue collar, white collar, and professional. Marital status might be coded as single, married, divorced, and widowed.

2.2 Binary outcomes

Binary dependent variables are extremely common in the social sciences. In these models dependent variable $y = 1$ when some events occurs and $y = 0$ if the event does not occur.

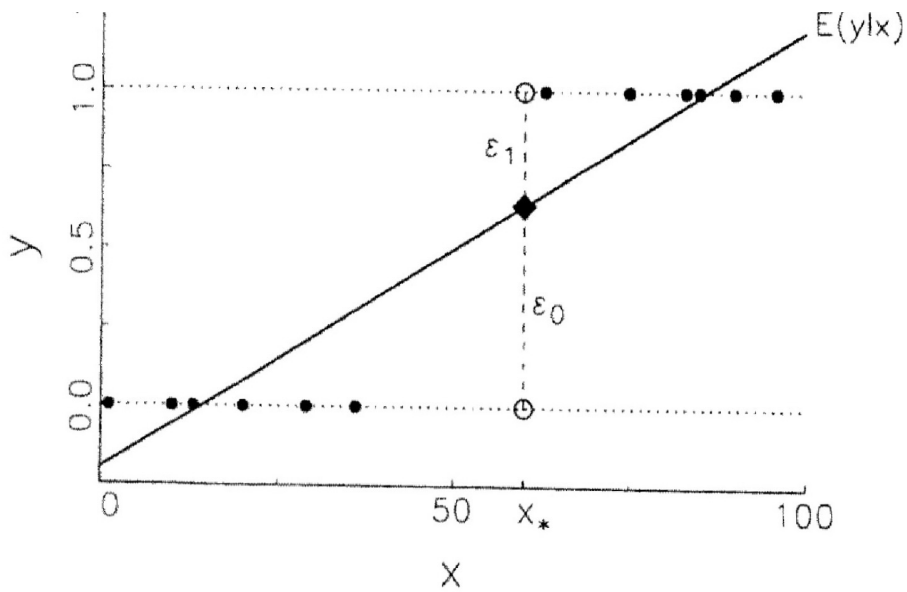
2.2.1 The linear probability model

The linear probability model is the regression model applied to a binary dependent variable. If we have a single explanatory variable, the model can be written as

$$y_n = \alpha + \beta x_n + \epsilon_n \ ,$$

which is plotted in Figure 1.

Figure 1: The linear probability model



Source: Long (1997)

The conditional expectation of y given x, $E(y|x) = \alpha + \beta x_n$ is the solid line in Figure 1. Observations are points at $y = 0$ and $y = 1$.

As y is a binary random variable, the unconditional expectation of y is the probability that the events occurs:

$$E(y_n) = [1 \cdot Pr(y_n = 1)] + [0 \cdot Pr(y_n = 0)] = Pr(y_n = 1) \ .$$

For the regression model, we are taking conditional expectations:

$$E(y_n|x_n) = [1 \cdot Pr(y_n = 1|x_n)] + [0 \cdot Pr(y_n = 0|x_n)] = Pr(y_n = 1|x_n) \ .$$

Therefore, the expected value of y given x is the probability that $y = 1$ given x. This allows us to rewrite the LPM as

$$Pr(y_n = 1|x_n) = \alpha + \beta x_n \ .$$

Having a binary outcome does not affect the interpretation of the parameters in the model. For a unit increase in x_n, the expected change in the probability of an event occurring is β. Since the model is linear, a unit change in x_n always results in the same change in the probability.

Problems with the LPM:

- *Heteroskedasticity*: If a binary random variable has a mean μ, then its variance is $\mu(1-\mu)$. Therefore

$$\begin{aligned} Var(y_n|x_n) &= x_n\beta(1-x_n\beta) \\ &= Pr(y_n=1|x_n)(1-Pr(y_n=1|x_n)) \ . \end{aligned}$$

This implies that the variance of the errors depends on the value of x_n and is not constant for $\forall i$. Since the LPM is heteroskedastic, the OLS estimator of β is inefficient and the standard errors are biased, resulting in incorrect test statistic.

- *Normality:* If in LPM $y_n = x_n'\beta + \epsilon_n$ the dependent variable y can take only values of 1 and 0, then

$$\epsilon_n = \begin{cases} 1 - x_n'\beta & \text{if } y_n = i \\ -x_n'\beta & \text{if } y_n = 0. \end{cases}$$

Clearly, the errors cannot be normally distributed. Normality is not required for the OLS estimates to be unbiased.

- *Nonsensical predictions:* The LPM predicts values of y that are negative or greater than 1. This leads to nonsensical predictions for the probabilities.
- *Functional form:* Since the model is linear, a unit increase in x_k results in a constant change of β_k in the probability of an event, holding all other variable constant. The increase is the same regardless of the current value of x. This is unrealistic in many applications.

2.2.2 A latent variable model for binary variables

Suppose that there is an unobserved latent variable $y^\star$ ranging from $-\infty$ to ∞ that generates the observed binary response of y. Those who have larger values of $y^\star$ are observed as $y=1$, while those with smaller values of $y^\star$ are observed es $y=0$.

Consider a woman's labour force participation as the observed y. The idea of $y^\star$ is that there is an underlying propensity to work that generates the observed state. At same point, the propensity crosses a threshold that would result in decision to enter (or leave) the labor force.

The latent $y^\star$ is assumed to be linearly related to the observed x's through the structural model:

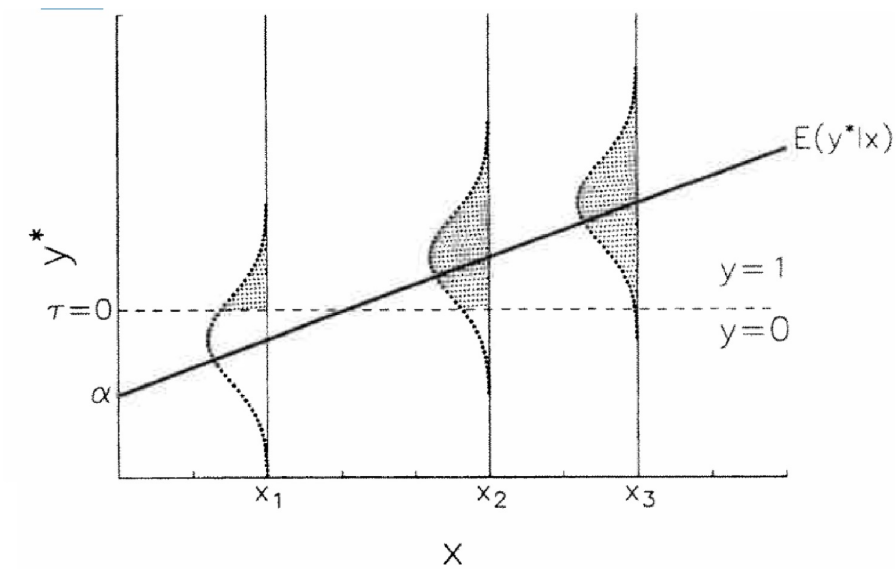
$$y_n^\star = \mathbf{x}_n'\beta + \epsilon_n \ . \tag{1}$$

The latent variable $y^\star$ is linked to the observed binary variable y by the measurement equation:

$$y_n = \begin{cases} 1 & \text{if } y_n^\star > \tau \\ 0 & \text{if } y_n^\star \leq \tau \ , \end{cases}$$

where τ is the *threshold* or *cutpoint*. The link between the latent variable $y^\star$ and observed y is illustrated in Figure 2. For example, at x_1 about 25% of y's equal 1, at x_2 nearly 90% are 1's, and at x_3 nearly all cases are 1's.

Figure 2: The distribution of $y^\star$ given x in the binary response model



Source: Long (1997)

Since $y^\star$ is continuous, this approach avoids problems encountered in LPM. Nevertheless $y^\star$ is unobserved and OLS estimation is not feasible. Different assumptions on error terms lead to ML estimation of models called *Logit* and *Probit*.

Another specific issue in these models is that the variance of the error term must be fixed because $y^\star$ is not observed and the model is not identified without an additional restriction. In the probit model we assume that $Var(\epsilon|\mathbf{x}) = 1$ and in the logit model $Var(\epsilon|\mathbf{x}) = \Pi^2/3$.

When ϵ in (1) is assumed to be normal with $E(\epsilon|x) = 0$ and $V(\epsilon|x) = 1$, the *pdf* is

$$\phi(\epsilon) = \frac{1}{\sqrt{2\pi}} exp(-\frac{\epsilon^2}{2})$$

and the cumulative distribution function is

$$\Phi(\epsilon) = \int_{-\infty}^{\epsilon} \frac{1}{\sqrt{2\pi}} exp(-\frac{t^2}{2})dt \ .$$

In this case the model is called *Probit*. In the *Logit* model, the errors ϵ in (1) are assumed to have a logistic distribution with mean 0 and variance $\pi^2/3$. This unusual variance is chosen because it results in a particulary simple equation for the *pdf*:

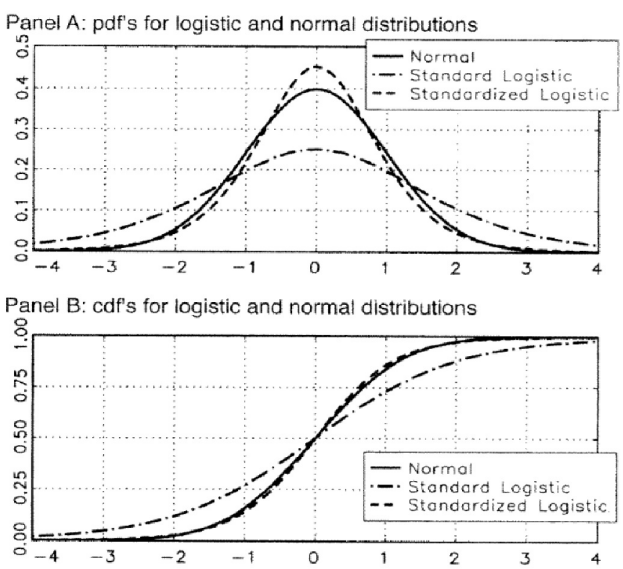
$$\lambda(\epsilon) = \frac{exp(\epsilon)}{(1 + exp(\epsilon))^2}$$

and even simpler equation for the *cdf*

$$\Lambda(\epsilon) = \frac{exp(\epsilon)}{1 + exp(\epsilon)} \ .$$

If we rescale the logistic distribution to have a unit variance, known as the standardized (not standard) logistic distribution, the logistic and normal *cdf*'s are nearly identical (Figure 3).

Figure 3: Normal and Logistic Distributions



Source: Long (1997)

By assuming a specific form for the distribution of ϵ, it is possible to compute the probability of $y = 1$ for

given x'. According to Figure 2

$$P(y_n = 1|\mathbf{x}_n) = P(\epsilon \leq \mathbf{x}_n'\beta|x_n) \ .$$

This is however the *cdf* of the error distribution evaluated at $\mathbf{x}_n'\beta$. Accordingly

$$P(y_n = 1|\mathbf{x}_n) = F(\mathbf{x}_n'\beta|\mathbf{x}_n) \ ,$$

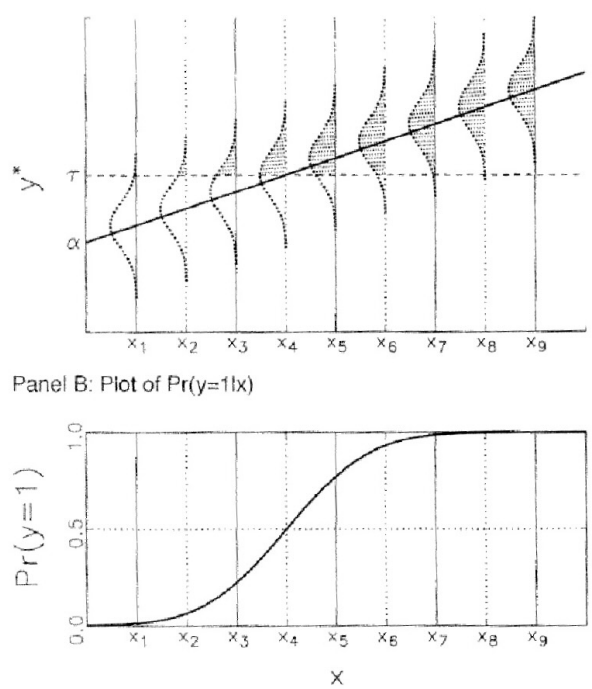
where F is the normal *cdf* Φ for the probit model and logistic *cdf* Λ for the logit model.

The interpretation of this model can be easily seen in the case of a single independent variable x:

$$P(y_n = 1|x_n) = F(\alpha + \beta x_n) \ .$$

As x_n increases by one unit, the argument of F increases by β units which causes its corresponding increase in F (Figure 4).

Figure 4: Plot of $y^\star$ and $Pr(y = 1|x)$ in the Binary Response Model



Source: Long (1997)

In specifying binary logit and probit model, we made three identifying assumption:

- $\tau = 0$
- $E(\epsilon) = 0$
- $Var(\epsilon_n|x_n) = 1 \quad \forall i$ in the probit model and $Var(\epsilon_n|x_n) = \pi^2/3 \quad \forall i$ in the logit model.

This is because our latent variable is unobserved and its mean and variance cannot be estimated. That is why the β's are unidentified without assumption about mean and variance of ϵ. In this sense the β's are arbitrary. Accordingly, the β's cannot be interpreted directly since they reflect both:

- the relationship between the $\mathbf{x}$'s and $y^\star$
- the identification assumptions.

While the specific value assumed for $Var(\epsilon|\mathbf{x})$ is arbitrary and affects β's it does not affect the quantity that is of fundamental interest, namely, the probability that an event occurred.

The Binary Regression Model can be also derived without appealing to underlying latent variable.

This is done by specifying a non-linear model relating the x's to the probability of an event. We can transform the probability $Pr(y = 1|\mathbf{x})$ into *odds*:

$$\frac{Pr(y = 1|\mathbf{x})}{Pr(y = 0|\mathbf{x})} = \frac{Pr(y = 1|\mathbf{x})}{1 - Pr(y = 1|\mathbf{x})} \ .$$

The odds indicate how often something (e.g., $y = 1$) happens relative to how often it does not happen (e.g., $y = 0$), and range from 0 when $Pr(y = 1|\mathbf{x}) = 0$ to ∞ when $Pr(y = 1|\mathbf{x}) = 1$. The log of the odds, known as the logit, ranges from $-\infty$ to ∞. This suggests a model that is linear in the logit:

$$ln\left[\frac{Pr(y = 1|\mathbf{x})}{1 - Pr(y = 1|\mathbf{x})}\right] = \mathbf{x}\beta \ .$$

This is, however, equivalent to the logit model derived above:

$$Pr(y = 1|\mathbf{x}) = \frac{exp(\mathbf{x}\beta)}{1 + exp(\mathbf{x}\beta)} \ .$$

2.2.3 ML estimation

Let us define p_n as

$$p_n = \begin{cases} Pr(y_n = 1|\mathbf{x}_n) & \text{if } y_n = 1 \text{ is observed} \\ 1 - Pr(y_n = 1|\mathbf{x}_n) & \text{if } y_n = 0 \text{ is observed} \end{cases},$$

where

$$P(y_n = 1|\mathbf{x}_n) = F(\mathbf{x}'\beta|\mathbf{x}) \ .$$

If the observations are independent, the likelihood equation is

$$\begin{aligned} L(\beta|\mathbf{y}, \mathbf{X}) &= \prod_{n=1}^{N} p_n \\ &= \prod_{y=1} P(y_n = 1|\mathbf{x}_n) \prod_{y=0} [1 - P(y_n = 1|\mathbf{x}_n)] \\ &= \prod_{y=1} F(x_n'\beta) \prod_{y=0} [1 - F(x_n'\beta)]) \ , \end{aligned}$$

where the index for multiplication indicates that the product is taken over those cases where $y = 1$ and $y = 0$ respectively.

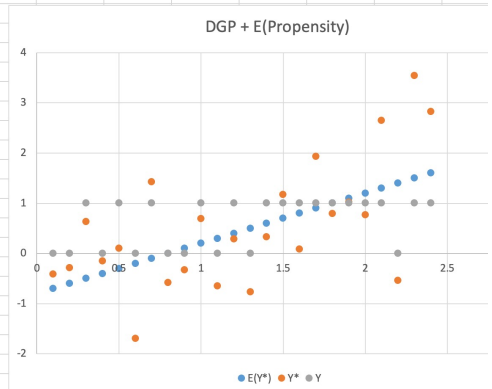
Taking logs, we obtain the log-likelihood equation:

$$\ln L(\beta|y, x) = \sum_{y=1} \ln F(x_n'\beta) + \sum_{y=0} \ln[1 - F(x_n'\beta)]) \ .$$

It can be shown that under conditions that are likely to apply in practice, the likelihood function is globally concave which ensures the uniqueness of the ML estimates.

Example in Excel: **Excel_BinaryLogitModelEstimationLogL.xlsx**

	A	B	C	D	E	F	G	H	I	J	K	L
1		Population parameters						Estimated parameters				
2		beta1	-0.80					beta1	-1.33			
3		beta2	1.00					beta2	1.59			
4											SumLog	
5										Prob of	-13.211439	
6	X	E(Y*)	epsilon	Y*	Y		hat(Y*)	P(Y=0)	P(Y=1)	chosen	Ind. Log-Lik	CDF
7	0.1	-0.7	0.287	-0.413	0		-1.172	0.764	0.236	0.764	-0.270	0.236
8	0.2	-0.6	0.317	-0.283	0		-1.013	0.734	0.266	0.734	-0.310	0.266
9	0.3	-0.5	1.133	0.633	1		-0.855	0.702	0.298	0.298	-1.209	0.298
10	0.4	-0.4	0.247	-0.153	0		-0.696	0.667	0.333	0.667	-0.405	0.333
11	0.5	-0.3	0.401	0.101	1		-0.537	0.631	0.369	0.369	-0.997	0.369
12	0.6	-0.2	-1.493	-1.693	0		-0.379	0.594	0.406	0.594	-0.522	0.406
13	0.7	-0.1	1.522	1.422	1		-0.220	0.555	0.445	0.445	-0.809	0.445
14	0.8	0	-0.582	-0.582	0		-0.061	0.515	0.485	0.515	-0.663	0.485
15	0.9	0.1	-0.425	-0.325	0		0.098	0.476	0.524	0.476	-0.743	0.524
16	1	0.2	0.489	0.689	1		0.256	0.436	0.564	0.564	-0.573	0.564
17	1.1	0.3	-0.946	-0.646	0		0.415	0.398	0.602	0.398	-0.922	0.602
18	1.2	0.4	-0.116	0.284	1		0.574	0.360	0.640	0.640	-0.447	0.640
19	1.3	0.5	-1.262	-0.762	0		0.732	0.325	0.675	0.325	-1.125	0.675
20	1.4	0.6	-0.271	0.329	1		0.891	0.291	0.709	0.709	-0.344	0.709
21	1.5	0.7	0.476	1.176	1		1.050	0.259	0.741	0.741	-0.300	0.741



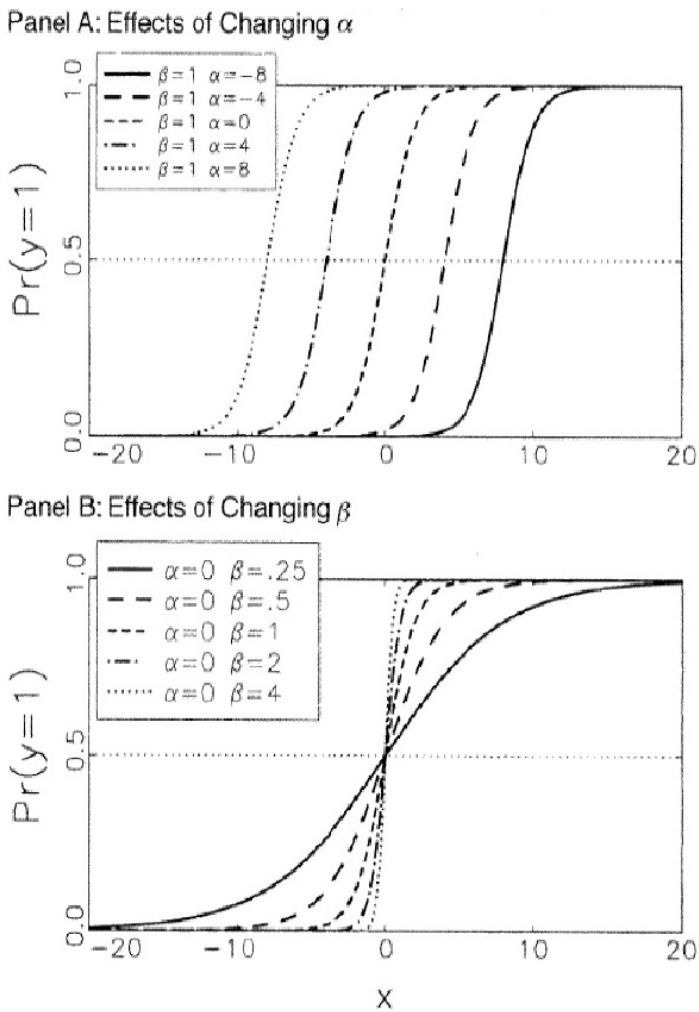
2.2.4 Interpretation

Since Binary Regression Model is non-linear, no single approach to interpretation fully describes the relationship between a variable and the outcome probability. You should search for an elegant and concise way to summarize your results.

- *The effects of the parameters*

Consider the Binary Regression Model $Pr(y = 1|x) = F(\alpha + \beta x)$. As α gets larger, the curve shifts to the left. The smaller the β, the more stretched out the curve. When the slope is negative, the curve is rotated $180°$ around $x = 0$.

Figure 5: Effects of changing the slope and intercept



Source: Long (1997)

- *Interpretation using predicted probabilities*

When there are more than two variables, it is no longer possible to plot the entire probability surface and a decision must be made regrading which probabilities to compute and how to present them. It is useful to compute:

1) *The range of probabilities*

$$\begin{aligned} \min\ \hat{P}r(y=1|\mathbf{x}) &= \min_n\ F(\mathbf{x}_n'\hat{\beta}) \\ \max\ \hat{P}r(y=1|\mathbf{x}) &= \max_n\ F(\mathbf{x}_n'\hat{\beta}) \end{aligned}$$

2) *The effects of each variable on the predicted probability*

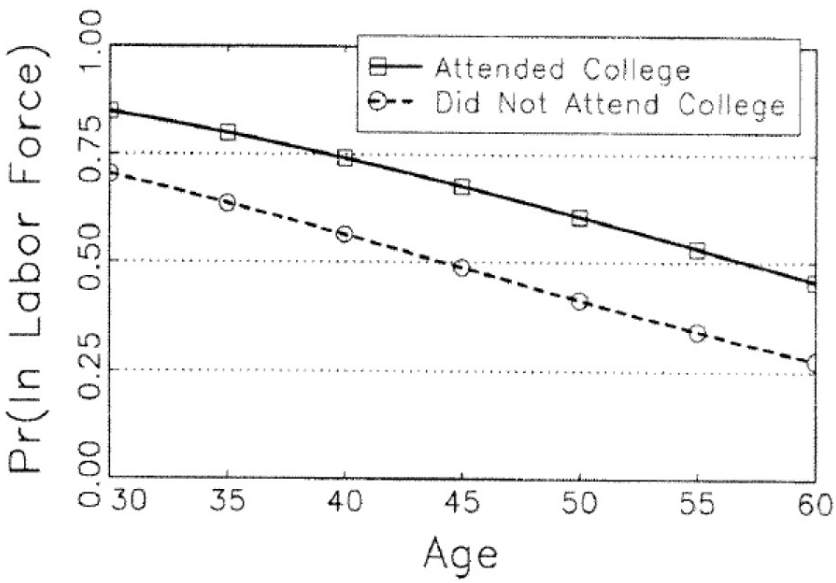
$$Pr(y = 1|\bar{\mathbf{x}}, \max\ x_k) - Pr(y = 1|\bar{\mathbf{x}}, \min\ x_k)$$

If one of x is a dummy variable, $\bar{x}$ has no interpretation. Values 0 or 1 must be used.

3) *Plotting probabilities over the range of a variable*

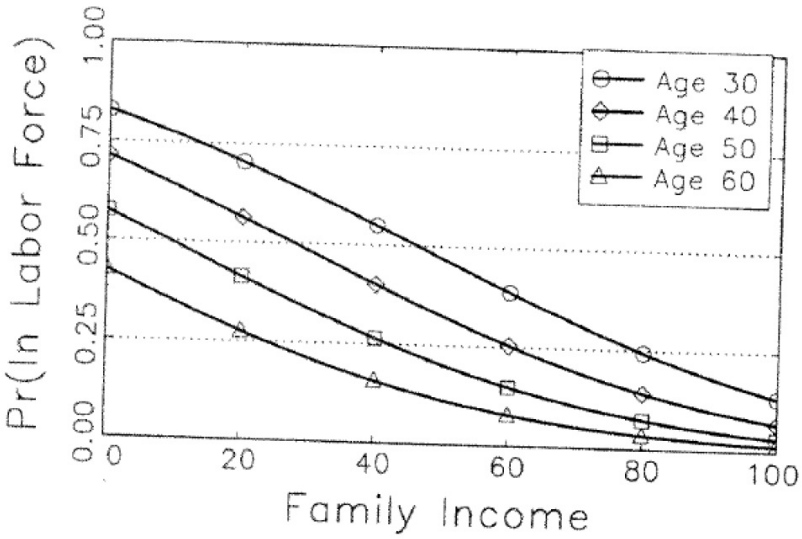
When there are more than two independent variables, we must examine the effects of one or two variables while the remaining variables are held constant (at mean value or value convenient for the interpretation).

Figure 6: Example of plot of probability of labor force participation by age and wife's education (family income is set to mean)



Source: Long (1997)

Figure 7: Example of plot of probability of labour force participation by age and family income for women without some college education



Source: Long (1997)

- *The partial change in* $Pr(y = 1|x)$

In our Binary regression Model

$$Pr(y = 1|\mathbf{x}) \quad = \quad F(\mathbf{x}'\beta)$$

where F is either the *cdf* for the normal distribution or the *cdf* Λ for the logistic distribution. The corresponding *pdf* is indicated as *f*. The partial change in the probability, also called the marginal effect, is defined as

$$\frac{\partial Pr(y = 1|\mathbf{x})}{\partial x_k} \quad = \quad \frac{\partial F(\mathbf{x}'\beta)}{\partial x_k} = \frac{dF(\mathbf{x}'\beta)}{d\,\mathbf{x}'\beta}\frac{\partial \mathbf{x}'\beta}{\partial x_k} = f(\mathbf{x}'\beta)\beta_k \ .$$

The sign of the marginal effect is determined by β_k since $f(\mathbf{x}'\beta)$ is always positive. The magnitude, however, depends on $f(\mathbf{x}'\beta)$ and is not directly expressed by β_k.

The ratio of marginal effects for x_k and x_ℓ is

$$\frac{\frac{\partial Pr(y=1|\mathbf{x})}{\partial x_k}}{\frac{\partial Pr(y=1|\mathbf{x})}{\partial x_l}} = \frac{f(\mathbf{x}'\beta)\beta_k}{f(\mathbf{x}'\beta)\beta_\ell} = \frac{\beta_k}{\beta_\ell}$$

While the β's are only identified up to a scale factor, their ratio is identified.

Since the value of the marginal effect depends on the levels of all variables, we must decide on which values of the variables to use when computing the effect. There are many alternatives. For example, we can use the average over all observations

$$\text{mean} \quad \frac{\partial Pr(y=1|\mathbf{x})}{\partial x_k} = \frac{1}{N}\sum_{n=1}^{N} f(x_n'\beta)\beta_k$$

or the marginal effect at the mean of the independent variables

$$\frac{\partial Pr(y=1|\bar{\mathbf{x}})}{\partial x_k} = f(\bar{\mathbf{x}}'\beta)\beta_k \ .$$

The second alternative is frequently included in econometric software but is limited. First, given the nonlinearity of the model, it is difficult to translate the marginal effect into the change in the predicted probability that will occur if there is a discrete change in x_k. Second, x_k might not correspond to any observed values in the population and is an inappropriate value for binary independent variables.

- *Discrete change in* $Pr(y=1|\mathbf{x})$

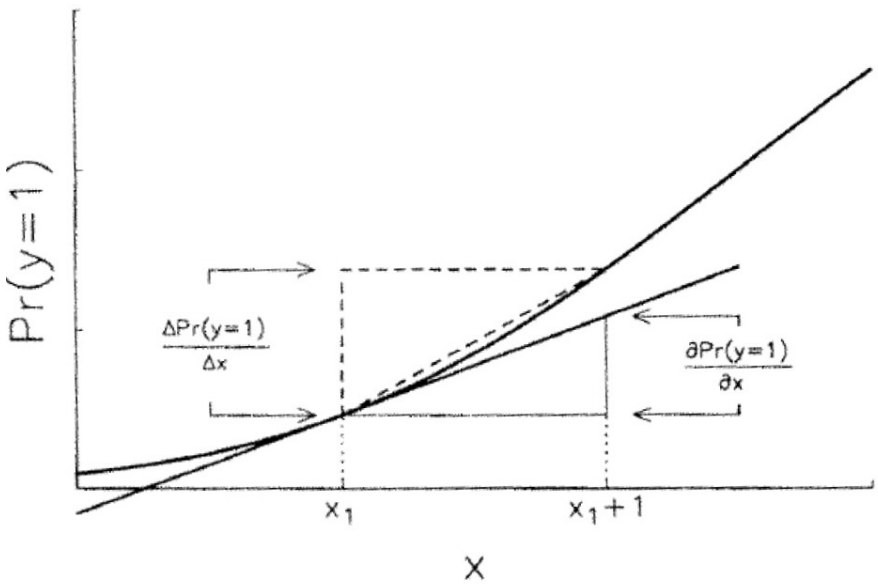
The discrete change in the probability for a change of δ in x_k equals

$$\frac{\triangle Pr(y=1|\mathbf{x})}{\triangle x_k} = Pr(y=1|\mathbf{x}, x_k+\delta) - Pr(y=1|\mathbf{x}, x_k) \ .$$

The discrete change can be interpreted as: for a change in the variable x_k from x_k to $x_k+\delta$, the predicted probability of an event changes by $\triangle Pr(y=1|\mathbf{x})/\triangle x_k$ holding all other variables constant.

It is essential to understand that the partial change does not equal the discrete change (Figure 8) except in the limit as δ becomes infinitely small (which is by definition the partial change).

Figure 8: Partial change versus discrete change in non-linear models



Source: Long (1997)

The amount of discrete change in the probability for a change in x_k depends on

- the amount of change in x_k
- the starting value of x_k
- the values of all other variables

There are many alternatives:

(1) *A unit change in x_k can be computed as*

$$\frac{\triangle Pr(y=1|\bar{\mathbf{x}})}{\triangle x_k} = Pr(y=1|\bar{\mathbf{x}}, \bar{x}_k + 1) - Pr(y=1|\bar{\mathbf{x}}, \bar{x}_k)$$

or

$$\frac{\triangle Pr(y=1|\bar{\mathbf{x}})}{\triangle x_k} = Pr(y=1|\bar{\mathbf{x}}, \bar{x}_k + \frac{1}{2}) - Pr(y=1|\bar{\mathbf{x}}, \bar{x}_k - \frac{1}{2}) \ .$$

(2) *Standard deviation change is*

$$\frac{\triangle Pr(y=1|\bar{\mathbf{x}})}{\triangle x_k} = Pr(y=1|\bar{\mathbf{x}}, \bar{x}_k + \frac{s_k}{2}) - Pr(y=1|\bar{\mathbf{x}}, \bar{x}_k - \frac{s_k}{2}) \ ,$$

where s_k is the standard deviation of x_k. - *Change from* 0 *to* 1 *for dummy variables*

$$\frac{\triangle Pr(y=1|\bar{\mathbf{x}})}{\triangle x_k} = Pr(y=1|\bar{\mathbf{x}}, x_k=1) - Pr(y=1|\bar{\mathbf{x}}, x_k=0) \ .$$

(3) *Interpretation using odds ratios*

The *logit* model can be written as the log-linear model:

$$ln\Omega(\mathbf{x}) = \mathbf{x}\beta \ ,$$

where

$$\Omega(\mathbf{x}) = \frac{Pr(y=1|\mathbf{x})}{Pr(y=0|\mathbf{x})} = \frac{Pr(y=1|\mathbf{x})}{1-Pr(y=1|\mathbf{x})}$$

is the odds of the event given $\mathbf{x}$ and $ln\,\Omega(\mathbf{x})$ is the log of the odds, known as the *logit*. Consequently

$$\frac{\partial ln\,\Omega(\mathbf{x})}{\partial x_k} = \beta_k \ .$$

Therefore, for a unit change in x_k, we expect the logit to change by β_k, holding all other variables constant.

It can be shown that

$$\frac{\Omega(\mathbf{x}, x_k+\delta)}{\Omega(\mathbf{x}, x_k)} = exp(\beta_k\delta) \ .$$

Therefore, for $\delta = 1$ we have: for a unit change in x_k, the odds are expected to change by a factor of $exp(\beta_k)$, holding all other variables constant.

2.2.5 Goodness of fit

Several pseudo-R^2's for binary models have been defined. These formulas produce different values in models with categorical outcomes, and, consequently, are though of as distinct measures.

- *The percentage of explained variation*

Let us define $\hat{\pi} = \hat{Pr}(y|\mathbf{x})$. Then

$$R^2_{Efron} = 1 - \frac{\sum_n^N (y_n - \hat{\pi}_n)^2}{\sum_n^N (y_n - \bar{y})^2}$$

Another measure uses the log-likelihood for model M_α without regressors, thought of as the total sum of squares and log-likelihood of model M_β with regressors as the residual sum of squares. Then

$$R^2_{McF} = 1 - \frac{ln\hat{L}(M_\beta)}{ln\hat{L}(M_\alpha)} \ .$$

As R^2_{McF} increases as new variables are added to the model, we can use adjusted $\bar{R}^2_{McF}$

$$\bar{R}^2_{McF} = 1 - \frac{ln\hat{L}(M_\beta) - K}{ln\hat{L}(M_\alpha)} \ ,$$

where K is number of parameters in the model.

- *The ratio of* $Var(y^\star)$ *and* $Var(\hat{y}^\star)$

For models defined in terms of a latent outcome according to $y^\star = \mathbf{x}'\beta + \epsilon$:

$$\bar{R}^2_{M\&Z} = \frac{\widehat{Var}(\hat{y}^\star)}{\widehat{Var}(y^\star)} = \frac{\widehat{Var}(\hat{y}^\star)}{\widehat{Var}(\hat{y}^\star) + Var(\epsilon)}$$

$\widehat{Var}(\hat{y}^\star)$ can be obtained by the use of $\mathbf{x}'\beta$ and the $Var(\epsilon)$ is set by the applied model (1 for Probit and $\pi^2/3$ in Logit)

- *A transformation of the likelihood ratio*

Let us define M_α as the model with just the intercept, and M_β as the model with regressors included, then

$$\bar{R}^2_{ML} = 1 - \left[\frac{L(M_\alpha)}{L(M_\beta)}\right]^{2/N} \tag{2}$$

- *Pseudo-*R^2 *using observed versus predicted values*

Let the observed y equal 0 or 1. The predicted probability that $y = 1$ is

$$\hat{p}_n = \hat{P}r(y = 1|\mathbf{x}_n) = F(\mathbf{x}_n'\hat{\beta}) \ ,$$

where F is the *cdf* for the normal distribution for *probit* and for the logisitic distribution for *logit*. Define the expected outcome $\hat{y}$ as

$$\hat{y}_n = \begin{cases} 0 & \text{if } \hat{\pi}_n \leq 0.5 \\ 1 & \text{if } \hat{\pi}_n > 0.5 \end{cases}$$

The *Count* R^2 is the proportion of correct predictions and is defined as

$$R^2_{Count} = \frac{1}{N} \sum_j n_{jj} \ ,$$

where the n_{jj}'s are the number of correct predictions for outcome j located on the diagonal cells in the following table:

	Predicted Outcome		
Observed Outcome	$\hat{y} = 1$	$\hat{y} = 0$	*Row Total*
$y = 1$	n_{11}: correct	n_{12}: incorrect	n_{1+}
$y = 0$	n_{21}: incorrect	n_{22}: correct	n_{2+}
Columns total	n_{+1}	n_{+2}	N

- *Information measures*

Akaike's Infromation Criterion (AIC) is defined as

$$AIC = -2\, ln\hat{L}(M_\beta) + 2\ K \ ,$$

where $\hat{L}(M_\beta)$ is the likelihood of the model and K is the number of parameters in the model. The quantity $-2\, ln\hat{L}(M_\beta)$ ranges from 0 to $+\infty$ with smaller values indicating a better fit. As the number of parameters K becomes larger, the numerator becomes larger ($2K$ is a penalty for increasing the number of parameters). All else being equal, smaller values suggest a better fitting model.

Bayesian Infromation Criterion (BIC) is defined as

$$BIC = -2\, ln\hat{L}(M_\beta) + K\ ln(N) \ .$$

BIC has been proposed as a measure to asses the overall fit of a model and to allow the comparison of both nested and non-nested models. Basically, BIC assesses whether the model fits the data sufficiently well to justify the number of parameters that are used. In a comparison of two models, the model with the smaller BIC is preferred.

2.2.6 Example: Labour force participation (Mroz's study)

```r
# Preliminaries
# Preliminaries
rm(list = ls(all = TRUE))             # Clear workspace
Sys.setenv(LANG = "en")               # Set language
setwd(".")                            # Set working
#-------------------------------------------------------------
# Example 1: Mroz's study

# Read Data Stored by 'Minitab', 'S', 'SAS', 'SPSS', 'Stata', ...
# read.dta() included in this library
if (!require("foreign")) {install.packages("foreign")}; library(foreign)
```

```
Loading required package: foreign
```

```r
# Analysis of Overdispersed Data
if (!require("aod")) {install.packages("aod")}; library(aod)
```

```
Loading required package: aod
```

```r
# Testing Linear Regression Models
# waldtest() included in this library
if (!require("lmtest")) {install.packages("lmtest")}; library(lmtest)
```

```
Loading required package: lmtest

Loading required package: zoo


Attaching package: 'zoo'

The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

```r
# Tools for Descriptive Statistics
# PseudoR2() included in this library
if (!require("DescTools")) {install.packages("DescTools")}; library(DescTools)
```

```
Loading required package: DescTools
```

```r
# Well-Formatted Regression and Summary Statistics Tables
if (!require("stargazer")) {install.packages("stargazer")}; library(stargazer)
```

```
Loading required package: stargazer


Please cite as:

 Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.

 R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
#
# Harrell Miscellaneous - Contains many functions useful for data analysis
if (!require("Hmisc")) {install.packages("Hmisc")}; library(Hmisc)
```

```
Loading required package: Hmisc
```

```
Attaching package: 'Hmisc'

The following objects are masked from 'package:DescTools':

    %nin%, Label, Mean, Quantile

The following objects are masked from 'package:base':

    format.pval, units
```

```r
# Marginal Effects for Model Objects
if (!require("margins")) {install.packages("margins")}; library(margins)
```

```
Loading required package: margins
```

```r
# ggplot2 is a system for declaratively creating graphics
if (!require("ggplot2")) {install.packages("ggplot2")}; library(margins)
```

```
Loading required package: ggplot2
```

```r
# Mroz's (1987) study of the labor force participation of women,  #
# using data from the 1976 Panel Study of Income Dynamics.        #
# The sample consists of 753 white, married women                 #
# between the ages of 30 and 60.                                  #
# The dependent variable lfp equals 1 if a woman is employed and  #
# else equals 0.                                                  #

data <- read.dta("https://raw.githubusercontent.com/discretechoice/Data/refs/heads/main/binlfp2.dta")

# Contains data from binlfp2.dta                                                  #
# obs:           753                          Data from 1976 PSID-T Mroz         #
# vars:            8                           30 Apr 2001 16:17                 #
# size:       10,542                           (_dta has notes)                  #
# ------------------------------------------------------------------------------ #
#   storage  display     value                                                    #
# variable name   type   format      label      variable label                    #
# ------------------------------------------------------------------------------ #
#   lfp             byte   %9.0g       lfplbl     Paid Labor Force: 1=yes 0=no   #
# k5              byte   %9.0g                  # kids < 6                       #
# k618            byte   %9.0g                  # kids 6-18                      #
# age             byte   %9.0g                  Wife's age in years              #
# wc              byte   %9.0g       collbl     Wife College: 1=yes 0=no         #
# hc              byte   %9.0g       collbl     Husband College: 1=yes 0=no      #
# lwg             float  %9.0g                  Log of wife's estimated wages    #
# inc             float  %9.0g                  Family income excluding wife's   #
# ------------------------------------------------------------------------------ #

# Variable |       Obs        Mean    Std. Dev.       Min        Max      #
# -------------+--------------------------------------------------------  #
# lfp |       753    .5683931    .4956295          0          1           #
# k5  |       753    .2377158     .523959          0          3           #
# k618|       753    1.353254    1.319874          0          8           #
# age |       753    42.53785    8.072574         30         60           #
# wc  |       753    .2815405    .4500494          0          1           #
# -------------+--------------------------------------------------------  #
#  hc |       753    .3917663    .4884694          0          1           #
# lwg |       753    1.097115    .5875564  -2.054124   3.218876           #
```

```r
# inc |       753    20.12897     11.6348  -.0290001         96           #
# summary(data)                                                          #
# head(data)                                                             #

# We estimate both the logit and probit models
# Logit
logit    <- glm(lfp ~ k5 + k618 + age + wc + hc + lwg + inc,
                data = data,
                family = "binomial"(link = "logit"))
summary(logit)
```

```
Call:
glm(formula = lfp ~ k5 + k618 + age + wc + hc + lwg + inc, family = binomial(link = "logit"),
    data = data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.182140   0.644375   4.938 7.88e-07 ***
k5          -1.462913   0.197001  -7.426 1.12e-13 ***
k618        -0.064571   0.068001  -0.950 0.342337
age         -0.062871   0.012783  -4.918 8.73e-07 ***
wcCollege    0.807274   0.229980   3.510 0.000448 ***
hcCollege    0.111734   0.206040   0.542 0.587618
lwg          0.604693   0.150818   4.009 6.09e-05 ***
inc         -0.034446   0.008208  -4.196 2.71e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  905.27  on 745  degrees of freedom
AIC: 921.27

Number of Fisher Scoring iterations: 4
```

```r
outreg1  <- summary(logit)$coefficients

# Probit
probit   <- glm(lfp ~ k5 + k618 + age + wc + hc + lwg + inc, data = data,
                family = binomial(link = "probit"))
summary(probit)
```

```
Call:
glm(formula = lfp ~ k5 + k618 + age + wc + hc + lwg + inc, family = binomial(link = "probit"),
    data = data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.918418   0.382356   5.017 5.24e-07 ***
k5          -0.874712   0.114423  -7.645 2.10e-14 ***
k618        -0.038595   0.040950  -0.942 0.345942
```

```
age         -0.037824   0.007605  -4.973 6.58e-07 ***
wcCollege    0.488310   0.136731   3.571 0.000355 ***
hcCollege    0.057172   0.124207   0.460 0.645306
lwg          0.365635   0.089992   4.063 4.85e-05 ***
inc         -0.020525   0.004852  -4.230 2.34e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  905.39  on 745  degrees of freedom
AIC: 921.39

Number of Fisher Scoring iterations: 4
```

```r
outreg2  <- summary(probit)$coefficients

# Comparing estimates of Logit and Probit
cbind(outreg1[,1],outreg2[,1])
```

```
                   [,1]        [,2]
(Intercept)  3.18214046  1.91841753
k5          -1.46291304 -0.87471237
k618        -0.06457068 -0.03859517
age         -0.06287055 -0.03782350
wcCollege    0.80727378  0.48830956
hcCollege    0.11173357  0.05717160
lwg          0.60469312  0.36563478
inc         -0.03444643 -0.02052513
```

```r
# The estimated coefficients differ from logit to probit by a factor of about 1.7.  #
# For example, the ratio of the logit to probit coefficient for k5 is 1.67 and for  #
# inc is 1.68. This illustrates how the magnitudes of the coefficients are          #
# affected by the assumed Var(e). The exception to the ratio of 1.7                 #
# is the coefficient for hc. Values of the z-tests are quite similar since they     #
# are not affected by the assumed Var(e) nevertheless they are not exactly the same #
# because the two models assume different distributions of the errors.              #


# Hypothesis tests of regression coefficients can be conducted with the  #
# z-statistics in the estimation output, with test for Wald tests of     #
# simple and complex hypotheses, and with lrtest for the corresponding   #
# likelihood-ratio tests                                                  #

wald.test(b      = coef(probit),
          Sigma  = vcov(probit),
          Terms  = 2)
```

```
Wald test:
----------

Chi-squared test:
X2 = 58.4, df = 1, P(> X2) = 2.1e-14
```

```r
# The effect of having young children on the probability of entering the      #
# labor force is significant at the .01 level (chi^2 = 58.4, df = 1, p < .01) #
# The value of a chi-squared test with 1 degree of freedom is identical       #
# to the square of the corresponding z-test.                                  #
print(58.0^0.5)
```

```
[1] 7.615773
```

```r
# We can test the same H_0 by LR test
# Full - Non restricted model
non.restricted.model <- glm(lfp ~ k5 + age + k618 + wc + hc + lwg + inc,
                            data   = data,
                            family = "binomial"(link = "logit"))
summary(non.restricted.model)
```

```
Call:
glm(formula = lfp ~ k5 + age + k618 + wc + hc + lwg + inc, family = binomial(link = "logit"),
    data = data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.182140   0.644375   4.938 7.88e-07 ***
k5          -1.462913   0.197001  -7.426 1.12e-13 ***
age         -0.062871   0.012783  -4.918 8.73e-07 ***
k618        -0.064571   0.068001  -0.950 0.342337
wcCollege    0.807274   0.229980   3.510 0.000448 ***
hcCollege    0.111734   0.206040   0.542 0.587618
lwg          0.604693   0.150818   4.009 6.09e-05 ***
inc         -0.034446   0.008208  -4.196 2.71e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  905.27  on 745  degrees of freedom
AIC: 921.27

Number of Fisher Scoring iterations: 4
```

```r
# Restricted model #
restricted.model  <- glm(lfp ~ k618 + age + wc + hc + lwg + inc,
                         data   = data,
                         family = "binomial"(link = "logit"))
summary(restricted.model)
```

```
Call:
glm(formula = lfp ~ k618 + age + wc + hc + lwg + inc, family = binomial(link = "logit"),
    data = data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.818452   0.528746   1.548  0.12164
```

```
k618        -0.001976   0.063656  -0.031  0.97523
age         -0.017112   0.010596  -1.615  0.10631
wcCollege    0.652128   0.215781   3.022  0.00251 **
hcCollege    0.028556   0.195493   0.146  0.88386
lwg          0.615340   0.145787   4.221 2.43e-05 ***
inc         -0.032787   0.007654  -4.284 1.84e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  971.75  on 746  degrees of freedom
AIC: 985.75

Number of Fisher Scoring iterations: 4
```

```r
# LR test #
lrtest.default(non.restricted.model,restricted.model)
```

```
Likelihood ratio test

Model 1: lfp ~ k5 + age + k618 + wc + hc + lwg + inc
Model 2: lfp ~ k618 + age + wc + hc + lwg + inc
  #Df  LogLik Df  Chisq Pr(>Chisq)
1   8 -452.63
2   7 -485.88 -1 66.484  3.527e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# The effect of having young children is significant at the .01 level
# (LR = 66.48, df = 1, p < .01).

# To test that the effect of the wife attending college and of the husband
# attending college on labor force participation are both equal to 0,
# H0: ?wc = ?hc = 0, we estimate the full model and then apply Wald test

wald.test(b     = coef(non.restricted.model),
          Sigma = vcov(non.restricted.model),
          Terms = 5:6)
```

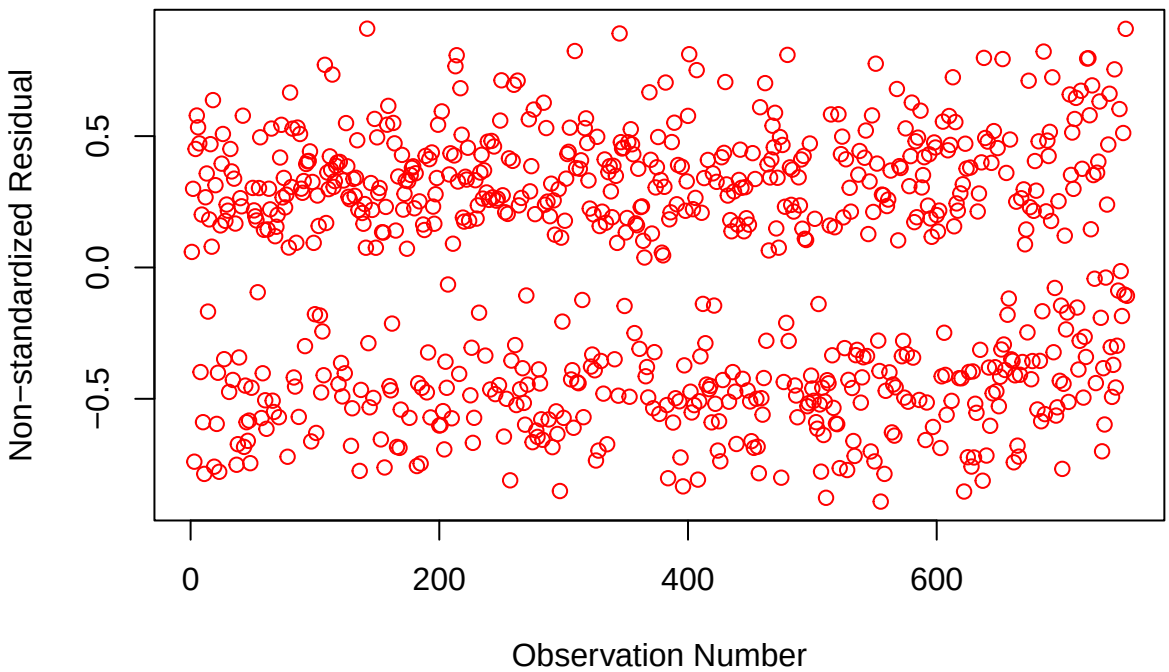
```
Wald test:
----------

Chi-squared test:
X2 = 17.7, df = 2, P(> X2) = 0.00015
```

```r
# ( 1)  [lfp]hc = 0                                                    #
# ( 2)  [lfp]wc = 0                                                    #
#                                                                      #
# chi2(  2) =   17.66                                                  #
# Prob > chi2 =    0.0001                                              #
#                                                                      #
# The hypothesis that the effects of the husband's and the wife's education  #
# are simultaneously                                                   #
# equal to zero can be rejected at the .01 level (W = 17.66, df = 2, p < .01). #
```

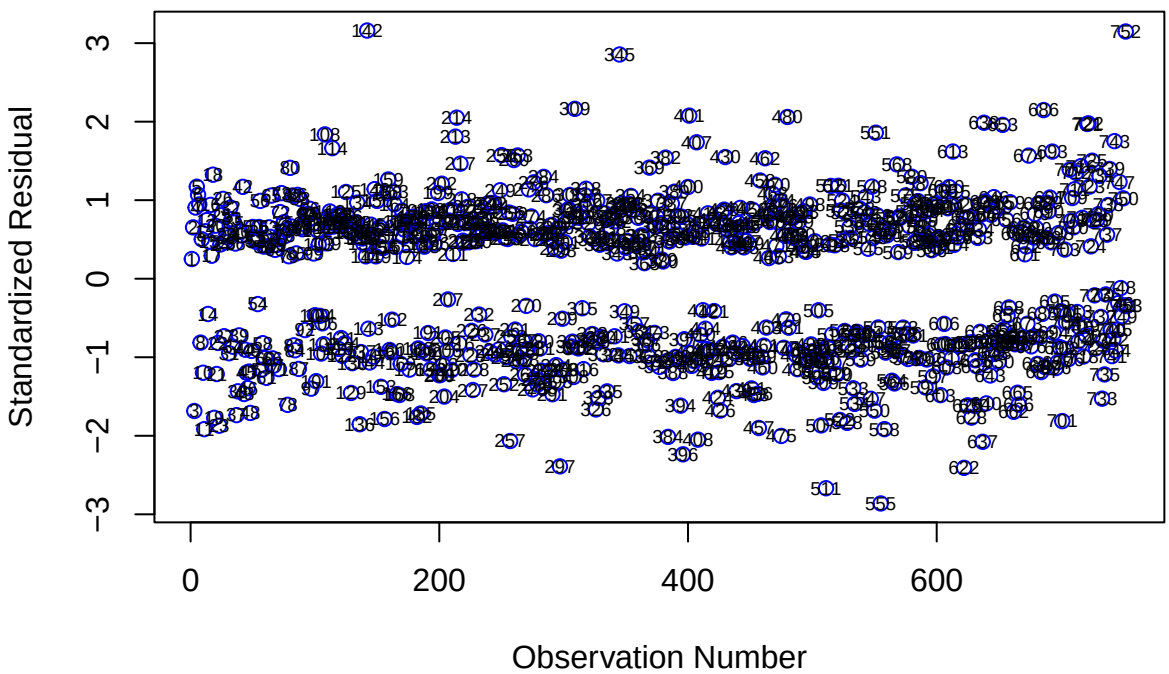
```r
# To draw some plots against for example "income", #
# Let us sort the data set                          #
#                                                   #
sorted.data  <- data[order(data$inc),]
logit        <- glm(lfp ~ k5 + k618 + age + wc + hc + lwg + inc,
                    data   = sorted.data,
                    family = "binomial"(link = "logit"))
summary(logit)
```

```
Call:
glm(formula = lfp ~ k5 + k618 + age + wc + hc + lwg + inc, family = binomial(link = "logit"),
    data = sorted.data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.182140   0.644375   4.938 7.88e-07 ***
k5          -1.462913   0.197001  -7.426 1.12e-13 ***
k618        -0.064571   0.068001  -0.950 0.342337
age         -0.062871   0.012783  -4.918 8.73e-07 ***
wcCollege    0.807274   0.229980   3.510 0.000448 ***
hcCollege    0.111734   0.206040   0.542 0.587618
lwg          0.604693   0.150818   4.009 6.09e-05 ***
inc         -0.034446   0.008208  -4.196 2.71e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  905.27  on 745  degrees of freedom
AIC: 921.27

Number of Fisher Scoring iterations: 4
```

```r
lfp           <- as.numeric(sorted.data$lfp == "inLF")
lfp.fitted    <- logit$fitted.values
rsntd         <- lfp-lfp.fitted
rstd          <- residuals.glm(logit, type = "pearson")
index         <- c(1:length(lfp))

plot(rsntd ~ index, xlab = "Observation Number"       ,
                    ylab = "Non-standardized Residual",
                    col  = "red")
```




```r
plot(rstd ~ index, xlab = "Observation Number"    ,
                   ylab = "Standardized Residual",
                   col  = "blue")

# We can instruct graph to use the observation number to #
# label each point in our plot                            #
plot(rstd ~ index, xlab = "Observation Number",
                   ylab = "Standardized Residual",
                   col  = "blue")
text(rstd ~ index, labels = index, cex = 0.6)
```



```r
# Outlier with number 142 in the sorted data base
sorted.data[142,]
```

```
     lfp k5 k618 age    wc    hc       lwg  inc
534 inLF  1    2  36 NoCol NoCol -2.054124 11.2
```

```r
# individuals with standardized residual greater than 2.5
rstd_del <- rstd[rstd > 2.5 | rstd < -2.5]
rstd_del
```

```
      534       662        60       252       635
 3.160038  2.855095 -2.668006 -2.861624  3.147970
```

```r
# There is no hard-and-fast rule for what counts as a "large" residual.   #
# One way to search for problematic residuals is to sort the residuals    #
# by the value of a variable that you think may be a problem for the      #
# model.                                                                   #
#                                                                          #
# In our example, we sorted the data by income before plotting.            #
# If this variable was primarily responsible for the lack of fit of some  #
# observations, the plot would show a disproportionate number of cases    #
# with large residuals among either the low income or high income         #
# observations in our model. However, this does not appear to be          #
# the case for these data                                                  #
#                                                                          #

#                                                                          #
# Goodness of fit measures                                                 #
#                                                                          #
# To illustrate the use of scalar measures of fit, consider two models.   #
#                                                                          #
# M1 contains our original specification of independent variables:        #
# k5, k618, age, wc, hc, lwg, and inc.                                     #
#                                                                          #
# M2 drops the variables k618 (not significant), hc (not significant)     #
# and lwg (significant) and adds (non-significant) agesq                  #
# which is the square of age.                                              #
#                                                                          #
# These models are estimated and measures of fit are computed              #
#                                                                          #
# Since the models are not nested, their comparision                       #
# is not straightforward. What do the fit statistics show?                 #

# Model 1
M1          <- glm(lfp ~ k5 + k618 + age + wc + hc + lwg + inc,
                   data   = sorted.data,
                   family = "binomial"(link = "logit"))
summary(M1)
```

```
Call:
glm(formula = lfp ~ k5 + k618 + age + wc + hc + lwg + inc, family = binomial(link = "logit"),
    data = sorted.data)

Coefficients:
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.182140   0.644375   4.938 7.88e-07 ***
k5          -1.462913   0.197001  -7.426 1.12e-13 ***
k618        -0.064571   0.068001  -0.950 0.342337
age         -0.062871   0.012783  -4.918 8.73e-07 ***
wcCollege    0.807274   0.229980   3.510 0.000448 ***
hcCollege    0.111734   0.206040   0.542 0.587618
lwg          0.604693   0.150818   4.009 6.09e-05 ***
inc         -0.034446   0.008208  -4.196 2.71e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  905.27  on 745  degrees of freedom
AIC: 921.27

Number of Fisher Scoring iterations: 4
```

```r
fitstatM1   <- PseudoR2(M1, which = "all")
Stata_names <- c("Model"                 ,
                 "Intercept-only"        ,
                 "D(df=747/745/2)"       ,
                 "LR(df=5/7/-2)"         ,
                 "p-value"               ,
                 "McFadden"              ,
                 "McFadden(adjusted)"    ,
                 "McKelvey & Zavoina"    ,
                 "Cox-Snell/ML"          ,
                 "Cragg-Uhler/Nagelkerke",
                 "Efron"                 ,
                 "Tjur's D"              ,
                 "AIC"                   ,
                 "AIC divided by N"      ,
                 "BIC"                   )

M1_values   <- c(fitstatM1[12]                                  ,
                 fitstatM1[13]                                  ,
                 M1$deviance                                    ,
                 fitstatM1[14]                                  ,
                 pchisq(fitstatM1[14], 752-745, lower.tail=FALSE),
                 fitstatM1[1]                                   ,
                 fitstatM1[2]                                   ,
                 fitstatM1[8]                                   ,
                 fitstatM1[3]                                   ,
                 fitstatM1[4]                                   ,
                 fitstatM1[7]                                   ,
                 fitstatM1[9]                                   ,
                 fitstatM1[10]                                  ,
                 fitstatM1[10]/length(M1$residuals)             ,
                 fitstatM1[11]                                  )

# Model 2
```

```r
agesq     <- sorted.data$age^2
M2        <- glm(lfp ~ k5 + age  + wc + inc + agesq,
               data   = sorted.data,
               family = "binomial"(link = "logit"))
summary(M2)
```

```
Call:
glm(formula = lfp ~ k5 + age + wc + inc + agesq, family = binomial(link = "logit"),
    data = sorted.data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.979168   2.458098   0.398    0.690
k5          -1.379839   0.195468  -7.059 1.68e-12 ***
age          0.056882   0.114110   0.498    0.618
wcCollege    1.093673   0.198739   5.503 3.73e-08 ***
inc         -0.032318   0.007728  -4.182 2.89e-05 ***
agesq       -0.001293   0.001294  -0.999    0.318
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  923.31  on 747  degrees of freedom
AIC: 935.31

Number of Fisher Scoring iterations: 4
```

```r
fitstatM2 <- PseudoR2(M2, which = "all")
M2_values   <- c(fitstatM2[12]                                    ,
                 fitstatM2[13]                                    ,
                 M2$deviance                                      ,
                 fitstatM2[14]                                    ,
                 pchisq(fitstatM2[14], 752-745, lower.tail=FALSE),
                 fitstatM2[1]                                     ,
                 fitstatM2[2]                                     ,
                 fitstatM2[8]                                     ,
                 fitstatM2[3]                                     ,
                 fitstatM2[4]                                     ,
                 fitstatM2[7]                                     ,
                 fitstatM2[9]                                     ,
                 fitstatM2[10]                                    ,
                 fitstatM2[10]/length(M2$residuals)               ,
                 fitstatM2[11]                                    )

# Output
stargazer(M1                                        ,
          M2                                        ,
          column.labels = c("Model M1", "Model M2"),
          keep.stat=c("n")                          ,
          digits=4                                  ,
          type="text"                               )
```

```
=========================================
                 Dependent variable:
             ----------------------------
                        lfp
                Model M1      Model M2
                  (1)            (2)
-----------------------------------------
k5             -1.4629***    -1.3798***
                (0.1970)      (0.1955)

k618            -0.0646
                (0.0680)

age            -0.0629***      0.0569
                (0.0128)      (0.1141)

wcCollege      0.8073***      1.0937***
                (0.2300)      (0.1987)

hcCollege        0.1117
                (0.2060)

lwg            0.6047***
                (0.1508)

inc            -0.0344***    -0.0323***
                (0.0082)      (0.0077)

agesq                          -0.0013
                              (0.0013)

Constant       3.1821***       0.9792
                (0.6444)      (2.4581)

-----------------------------------------
Observations      753            753
=========================================
Note:         *p<0.1; **p<0.05; ***p<0.01
```

```r
output                   <- data.frame(round(M1_values,3)            ,
                                       round(M2_values, 3)           ,
                                       round((M2_values-M1_values),3))
rownames(output)         <- Stata_names
colnames(output)         <- c("Model M1", "Model M2", "Difference")

output_list_STATA        <- list(output[1:2,]  ,
                                 output[3:5,]  ,
                                 output[6:12,] ,
                                 output[13:15,])
names(output_list_STATA) <- c("Log-likelihood",
                              "Chi-square"    ,
                              "R2"            ,
                              "IC"            )
```

```r
output_list_STATA
```

```
$`Log-likelihood`
               Model M1 Model M2 Difference
Model          -452.633 -461.653      -9.02
Intercept-only -514.873 -514.873       0.00

$`Chi-square`
                Model M1 Model M2 Difference
D(df=747/745/2)  905.266  923.306      18.04
LR(df=5/7/-2)    124.480  106.441     -18.04
p-value            0.000    0.000       0.00

$R2
                       Model M1 Model M2 Difference
McFadden                  0.121    0.103     -0.018
McFadden(adjusted)        0.105    0.092     -0.014
McKelvey & Zavoina        0.217    0.182     -0.035
Cox-Snell/ML              0.152    0.132     -0.021
Cragg-Uhler/Nagelkerke    0.204    0.177     -0.028
Efron                     0.155    0.135     -0.020
Tjur's D                  0.154    0.134     -0.020

$IC
                 Model M1 Model M2 Difference
AIC               921.266  935.306     14.040
AIC divided by N    1.223    1.242      0.019
BIC               958.258  963.050      4.791
```

```r
#                                                                          #
# The values of the pseudo-R2s are slightly  larger for M1 as a significant     #
# variable was dropped and only a nonsignificant variable was added.            #
# If you take the pseudo-R2s as evidence for the "best" model, there is some    #
# evidence preferring M1. Second, the AIC and BIC statistics is smaller for M1, #
# which provides support for that model.                                        #


# Predicted probabilities #
logit   <- glm(lfp ~ age + hc + inc + k5 + k618 + lwg + wc,
               data   = sorted.data,
               family = "binomial"(link = "logit"))

summary(logit)
```

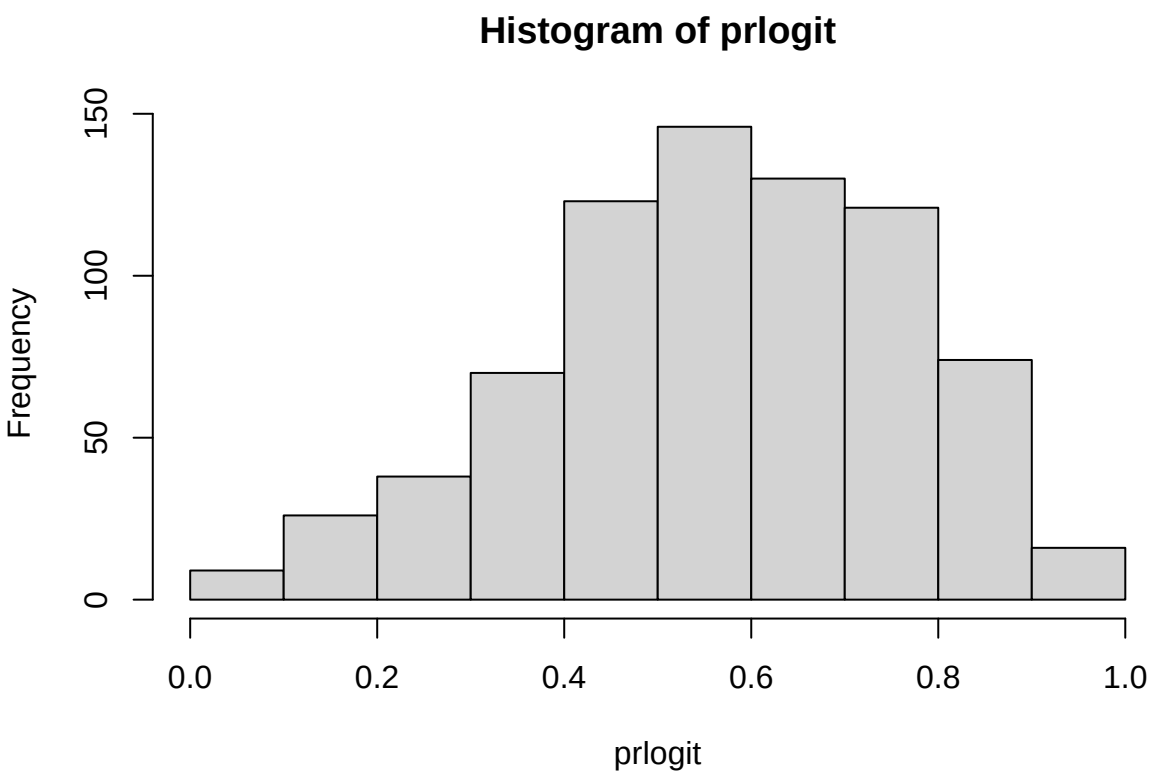
```
Call:
glm(formula = lfp ~ age + hc + inc + k5 + k618 + lwg + wc, family = binomial(link = "logit"),
    data = sorted.data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.182140   0.644375   4.938 7.88e-07 ***
age         -0.062871   0.012783  -4.918 8.73e-07 ***
hcCollege    0.111734   0.206040   0.542 0.587618
inc         -0.034446   0.008208  -4.196 2.71e-05 ***
```

```
k5          -1.462913   0.197001  -7.426 1.12e-13 ***
k618        -0.064571   0.068001  -0.950 0.342337
lwg          0.604693   0.150818   4.009 6.09e-05 ***
wcCollege    0.807274   0.229980   3.510 0.000448 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  905.27  on 745  degrees of freedom
AIC: 921.27

Number of Fisher Scoring iterations: 4
```

```r
prlogit <- logit$fitted.values
summary(prlogit)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.01399 0.44080 0.57706 0.56839 0.72133 0.96212
```

```r
hist(prlogit)
```




```r
# The plot clearly shows that the predicted probabilities for individual  #
# observations span almost the entire range from 0 to 1, but that roughly #
# two-thirds of the observations have predicted probabilities             #
# between .40 and .80.                                                    #
# we can compute predicted probabilities of labor force participation     #
# for women in different types of families                                #

# Family type:                                                  #
```

```r
# young, low income, low education families with young children #

newdata <- data.frame(
                      k5   = 2              ,
                      age  = 35             ,
                      k618 = mean(data$k618),
                      wc   = "NoCol"        ,
                      hc   = "NoCol"        ,
                      lwg  = mean(data$lwg) ,
                      inc  = 15
                     )

predict(logit, newdata, type = "response")
```

```
        1
0.1318369
```

```r
# Difference in probability to be in labour force between
# woman with and without kid (k5 = 0 or 1)
newdata2.nokid <- data.frame(
                             k5   = 0              ,
                             age  = mean(data$age) ,
                             k618 = mean(data$k618),
                             wc   = "College"      ,
                             hc   = "NoCol"        ,
                             lwg  = mean(data$lwg) ,
                             inc  = mean(data$inc)
                            )

newdata3.onekid <- data.frame(
                              k5    = 1              ,
                              age   = mean(data$age) ,
                              k618  = mean(data$k618),
                              wc    = "NoCol"        ,
                              hc    = "NoCol"        ,
                              lwg   = mean(data$lwg) ,
                              inc   = mean(data$inc)
                             )

# Difference between kid=1 and kid=0
c( predict(logit, newdata2.nokid , type = "response"),
   predict(logit, newdata3.onekid, type = "response"))
```

```
        1         1
0.7681158 0.2549309
```

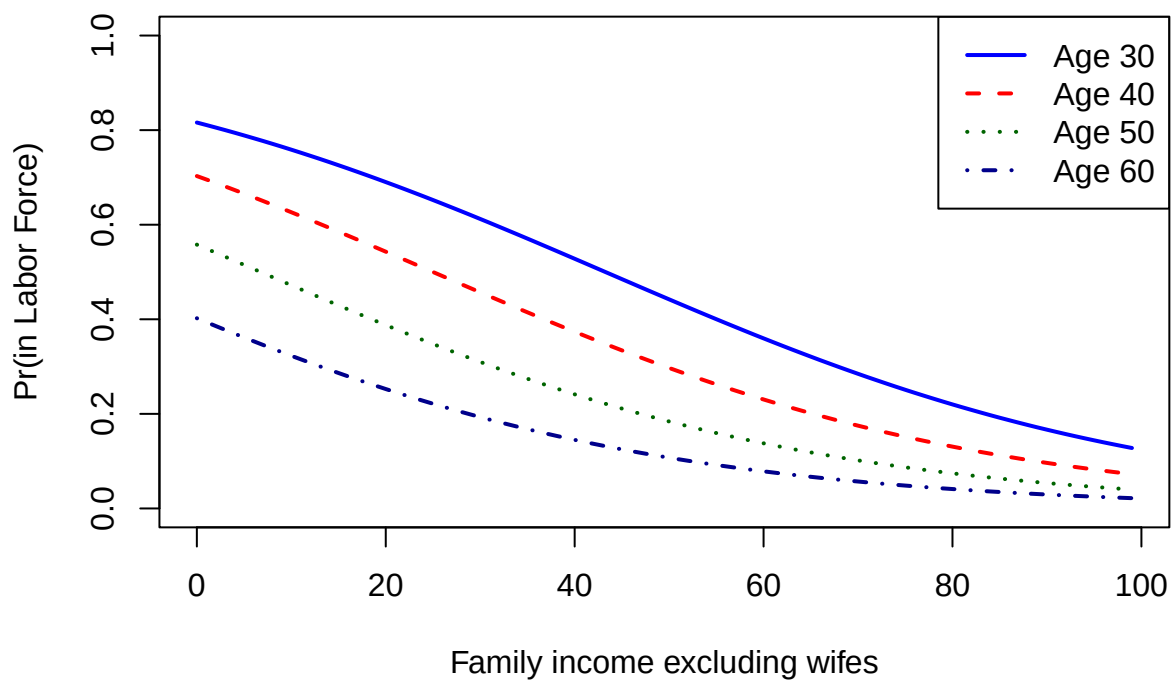
```r
# Diference in probability to be in labor force between #
# woman with and without College and number of kids < 6 #

table <- matrix(nrow = 4, ncol = 2)

for(j in 1:2)
  for(i in 1:4){
    if(j == 1){
      newdata3.onekid$k5 = i-1
```

```r
      table[i, j] <- predict(logit, newdata3.onekid, type = "response")}
    if(j == 2){
      newdata2.nokid$k5 = i-1
      table[i, j] <- predict(logit, newdata2.nokid, type = "response")}
  }

colnames(table) <- c("wcCollege = NO", "wcCollege = YES")
rownames(table) <- c("kids<6 = 0", "kids<6 = 1", "kids<6 = 2", "kids<6 = 3")
table
```

```
           wcCollege = NO wcCollege = YES
kids<6 = 0     0.59638591      0.76811582
kids<6 = 1     0.25493093      0.43408328
kids<6 = 2     0.07341365      0.15082775
kids<6 = 3     0.01801608      0.03950443
```

```r
# This table shows the strong effect on labour force participation #
# of having young children and how the effect differs according   #
# to the wife's education                                          #
# Now, we generate the predicted probabilities for those aged     #
# 30, 40, 50, 60                                                   #

newdata.age30 <- data.frame(
                            k5   = mean(data$k5)  ,
                            age  = 30             ,
                            k618 = mean(data$k618),
                            wc   = "NoCol"        ,
                            hc   = "NoCol"        ,
                            lwg  = mean(data$lwg) ,
                            inc  = mean(data$inc)
                            )

newdata.age40     <- newdata.age30
newdata.age50     <- newdata.age30
newdata.age60     <- newdata.age30
newdata.age40$age <- 40
newdata.age50$age <- 50
newdata.age60$age <- 60

p30          <- matrix(nrow=100, ncol=1)
p40          <- matrix(nrow=100, ncol=1)
p50          <- matrix(nrow=100, ncol=1)
p60          <- matrix(nrow=100, ncol=1)

for(i in 1:101){
                 newdata.age30$inc <-i-1
                 newdata.age40$inc <-i-1
                 newdata.age50$inc <-i-1
                 newdata.age60$inc <-i-1
                 p30[i-1]          <- predict(logit, newdata.age30,type="response")
                 p40[i-1]          <- predict(logit, newdata.age40,type="response")
                 p50[i-1]          <- predict(logit, newdata.age50,type="response")
                 p60[i-1]          <- predict(logit, newdata.age60,type="response")
               }
```

```r
# Plotting the predicted probabilities

x.data   <- 0:99
plot(x.data                                  ,
     p30                                     ,
     col  = "blue"                           ,
     type = "l"                              ,
     xlab = 'Family income excluding wifes',
     ylab = 'Pr(in Labor Force)'             ,
     ylim = c(0,1)                           ,
     lwd  = 2                                )
points(x.data, p40, col = "red"       , type = "l", lty = 2, lwd = 2)
points(x.data, p50, col = "dark green", type = "l", lty = 3, lwd = 2)
points(x.data, p60, col = "dark blue" , type = "l", lty = 4, lwd = 2)

legend("topright"                                          ,
       legend = c("Age 30", "Age 40", "Age 50", "Age 60"),
       col    = c("blue","red","dark green", "dark blue"),
       lty    = c(1,2,3,4)                                ,
       ncol   = 1                                         ,
       lwd    = 2                                         )
```




```r
# Marginal effects

data$wc.numeric                       <- 0
data$wc.numeric[data$wc == "College"] <- 1
data$hc.numeric                       <- 0
data$hc.numeric[data$hc == "College"] <- 1

logit   <- glm(lfp ~ age + hc.numeric + inc + k5 + k618 + lwg + wc.numeric  + lwg,
               data   = data,
               family = "binomial"(link = "logit"))
```

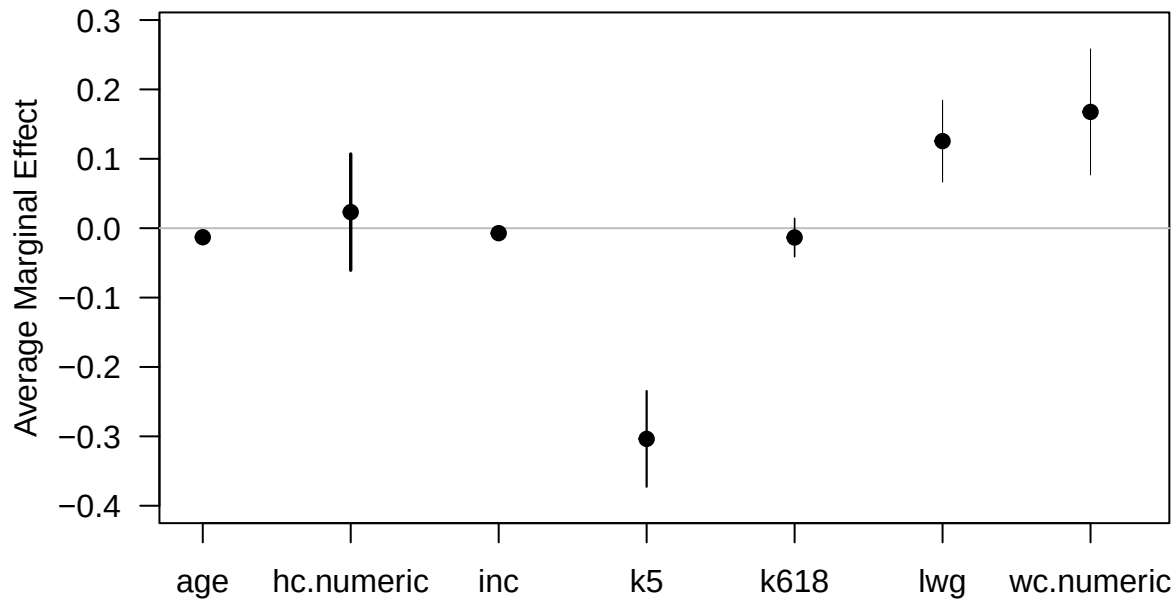
```r
summary(logit)
```

```
Call:
glm(formula = lfp ~ age + hc.numeric + inc + k5 + k618 + lwg +
    wc.numeric + lwg, family = binomial(link = "logit"), data = data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.182140   0.644375   4.938 7.88e-07 ***
age         -0.062871   0.012783  -4.918 8.73e-07 ***
hc.numeric   0.111734   0.206040   0.542 0.587618
inc         -0.034446   0.008208  -4.196 2.71e-05 ***
k5          -1.462913   0.197001  -7.426 1.12e-13 ***
k618        -0.064571   0.068001  -0.950 0.342337
lwg          0.604693   0.150818   4.009 6.09e-05 ***
wc.numeric   0.807274   0.229980   3.510 0.000448 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1029.75  on 752  degrees of freedom
Residual deviance:  905.27  on 745  degrees of freedom
AIC: 921.27

Number of Fisher Scoring iterations: 4
```

```r
average.marginal.change <- margins(logit)

# Average effects over all observations
summary(average.marginal.change)
```

```
     factor     AME     SE       z      p   lower   upper
        age -0.0131 0.0025 -5.2236 0.0000 -0.0179 -0.0082
 hc.numeric  0.0232 0.0427  0.5426 0.5874 -0.0606  0.1070
        inc -0.0072 0.0016 -4.3808 0.0000 -0.0103 -0.0040
         k5 -0.3037 0.0352 -8.6330 0.0000 -0.3726 -0.2347
       k618 -0.0134 0.0141 -0.9514 0.3414 -0.0410  0.0142
        lwg  0.1255 0.0301  4.1707 0.0000  0.0665  0.1845
 wc.numeric  0.1676 0.0464  3.6142 0.0003  0.0767  0.2584
```

```r
# names(average.marginal.change)
# mean(average.marginal.change$dydx_age)
 plot(average.marginal.change)
```




```r
# Marginal effect at the mean of the independent variables
summary(margins(logit, at=list(age        =  mean(data$age       ) ,
                               k5         =  mean(data$k5        ) ,
                               k618       =  mean(data$k618      ) ,
                               wc.numeric =  mean(data$wc.numeric) ,
                               hc.numeric =  mean(data$hc.numeric) ,
                               lwg        =  mean(data$lwg       ) ,
                               inc        =  mean(data$inc       )
                              )
                ))
```

```
     factor     age     k5   k618 wc.numeric hc.numeric    lwg     inc     AME
        age 42.5378 0.2377 1.3533     0.2815     0.3918 1.0971 20.1290 -0.0153
 hc.numeric 42.5378 0.2377 1.3533     0.2815     0.3918 1.0971 20.1290  0.0273
        inc 42.5378 0.2377 1.3533     0.2815     0.3918 1.0971 20.1290 -0.0084
         k5 42.5378 0.2377 1.3533     0.2815     0.3918 1.0971 20.1290 -0.3569
       k618 42.5378 0.2377 1.3533     0.2815     0.3918 1.0971 20.1290 -0.0158
        lwg 42.5378 0.2377 1.3533     0.2815     0.3918 1.0971 20.1290  0.1475
 wc.numeric 42.5378 0.2377 1.3533     0.2815     0.3918 1.0971 20.1290  0.1969
     SE       z      p   lower   upper
 0.0031 -4.9302 0.0000 -0.0214 -0.0092
 0.0503  0.5422 0.5877 -0.0713  0.1258
 0.0020 -4.1919 0.0000 -0.0123 -0.0045
 0.0482 -7.4024 0.0000 -0.4514 -0.2624
 0.0166 -0.9495 0.3424 -0.0483  0.0168
 0.0367  4.0146 0.0001  0.0755  0.2195
 0.0559  3.5200 0.0004  0.0873  0.3066
```

```r
# Interpretation of the mean of dummy variables: #
# wc.numeric =  mean(data$wc.numeric)            #
# hc.numeric =  mean(data$hc.numeric)            #
# is difficult                                   #

# Better representation is probably for a specific value #
# that have a clear interpretation:                      #
```

```r
summary(margins(logit, at=list(age        =  mean(data$age   ) ,
                               k5         =  0                 ,
                               k618       =  2                 ,
                               wc.numeric =  1                 ,
                               hc.numeric =  1                 ,
                               lwg        =  mean(data$lwg   ) ,
                               inc        =  mean(data$inc   )
                              )
               ))
```

```
     factor     age     k5   k618 wc.numeric hc.numeric    lwg     inc     AME
        age 42.5378 0.0000 2.0000     1.0000     1.0000 1.0971 20.1290 -0.0108
 hc.numeric 42.5378 0.0000 2.0000     1.0000     1.0000 1.0971 20.1290  0.0192
        inc 42.5378 0.0000 2.0000     1.0000     1.0000 1.0971 20.1290 -0.0059
         k5 42.5378 0.0000 2.0000     1.0000     1.0000 1.0971 20.1290 -0.2508
       k618 42.5378 0.0000 2.0000     1.0000     1.0000 1.0971 20.1290 -0.0111
        lwg 42.5378 0.0000 2.0000     1.0000     1.0000 1.0971 20.1290  0.1036
 wc.numeric 42.5378 0.0000 2.0000     1.0000     1.0000 1.0971 20.1290  0.1384
     SE       z      p   lower   upper
 0.0025 -4.3396 0.0000 -0.0156 -0.0059
 0.0348  0.5504 0.5821 -0.0490  0.0874
 0.0013 -4.3877 0.0000 -0.0085 -0.0033
 0.0359 -6.9754 0.0000 -0.3212 -0.1803
 0.0120 -0.9233 0.3558 -0.0346  0.0124
 0.0295  3.5126 0.0004  0.0458  0.1615
 0.0322  4.3013 0.0000  0.0753  0.2014
```

```r
# We have seen in previous plots that the relationship between age and the probability  #
# of being in the labour force was essentially linear for those who attend              #
# college. Accordingly, we can take the marginal computed by prchange (STATA),          #
# multiply it by 10 to get the amount of change over 10 years, and report that          #
# "For women who attend college and their husband also attended college                #
# a ten year increase in age decreases the                                              #
# probability of labour force participation by approximately .108,                       #
# holding other variables at their mean                                                 #
```

2.3 Ordinal Outcomes: Ordered Logit and Ordered Probit

When a variable is ordinal, its categories can be ranked from low to high, but the distances between adjacent categories are unknown. Many surveys have respondents indicating their income within ordered grouping, such as less than \$15,000, between \$15,000 and \$30,000, and so on. Likert scales on surveys ask respondents whether they strongly agree, agree, have no opinion, disagree, or strongly disagree with a statement.

Researchers often treat ordinal dependent variable as if they were interval. The dependent categories are numbered sequentially and the linear regression model is used. This involves the implicit assumption that the intervals between adjacent categories are equal.

2.3.1 A latent variable model for ordinal variables

The *ordered regression model* can be derived from a measurement model in which a latent variable $y^\star$ ranging from $-\infty$ to $+\infty$ is mapped to an observed variable y. The variable y is though of as providing incomplete information about an underlying $y^\star$ according to the measurement equation:

$$y_n = m \text{ if } \tau_{m-1} \leq y_n^\star < \tau_m \text{ for } m = 1 \text{ to } J \ .$$

The τ's are called *thresholds* or *cutpoints*. When $J = 2$ this measurement equation corresponds to a binary regression model analyzed in previous section.

To illustrate the measurement equation, consider the dependent variable y to be the following responses:

$$\hat{y}_n = \begin{cases} 1 & \text{Strongly Disagree (SD) if} & \tau_0 = -\infty \leq y_n^\star < \tau_1 \\ 2 & \text{Disagree (D) if} & \tau_1 \leq y_n^\star < \tau_2 \\ 3 & \text{Agree (A) if} & \tau_2 \leq y_n^\star < \tau_3 \\ 4 & \text{Strongly Agree (SA) if} & \tau_3 \leq y_n^\star < \tau_4 = +\infty \end{cases}$$

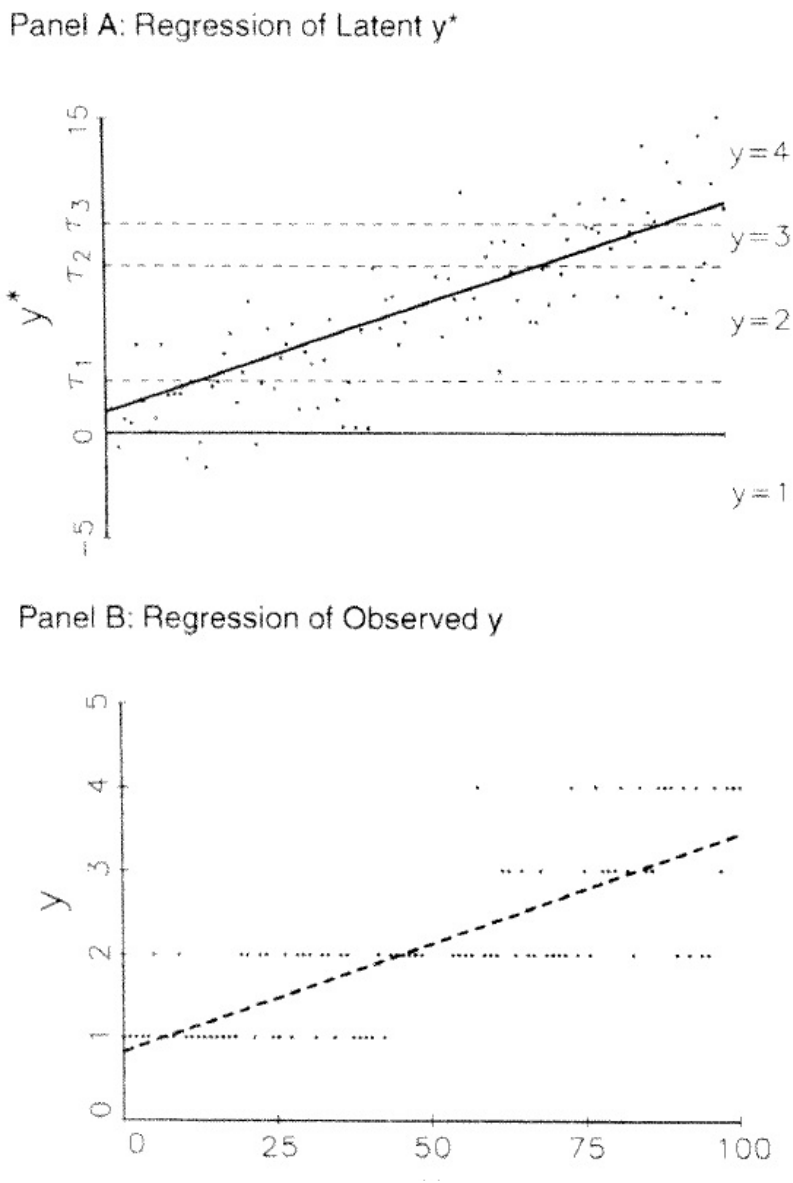
As with the binary response model, the structural model is

$$y_n^\star = \mathbf{x}_n' \beta + \epsilon_n \ ,$$

where $\mathbf{x}_n'$ is a row vector with a 1 in the first column for the intercept and the ith observation for x_k in column $k + 1$.

Figure 11 plots hypothetical values of $y^\star$ and y_n.

Figure 11: Regression of a latent variable $y^\star$ compared to the regression of the corresponding observed variable

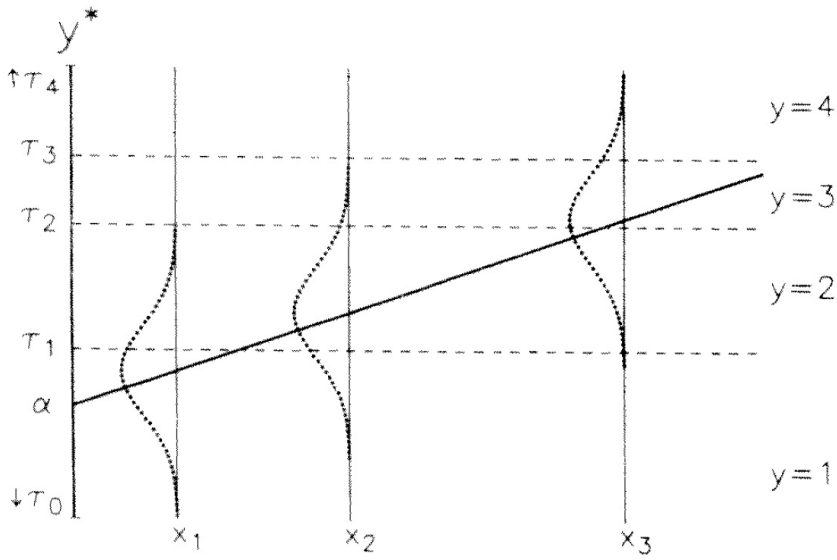


Source: Long (1997)

As with the binary regression model, the ML estimation can be used to estimate the regression of $y^\star$ on $\mathbf{x}$. To use ML, we must assume a specific form of the error distribution. Once again, we consider normal and logistic errors.

Figure 12 illustrates the distribution of $y^\star$ for three values of x. The errors are distributed either logistically or normally around the regression line $E(y^\star|x = \alpha + \beta x)$. The probability of outcome m corresponds to the area of the error distribution between the cutpoints τ_{m-1} and τ_m.

Figure 12: Distribution of $y^\star$ given x for ordered regression model



Source: Long (1997)

For a model with four observed outcomes, such as shown in Figure 12 the formulas for the ordered probit are

$$
\begin{aligned}
Pr(y_n = 1|x_n) &= \Phi(\tau_1 - \alpha - \beta x_n) \\
Pr(y_n = 2|x_n) &= \Phi(\tau_2 - \alpha - \beta x_n) - \Phi(\tau_1 - \alpha - \beta x_n) \\
Pr(y_n = 3|x_n) &= \Phi(\tau_3 - \alpha - \beta x_n) - \Phi(\tau_2 - \alpha - \beta x_n) \\
Pr(y_n = 4|x_n) &= 1 - \Phi(\tau_3 - \alpha - \beta x_n)
\end{aligned}
$$

Since $y^\star$ is latent, its mean and variance cannot be estimated. The variance is identified by assuming that $Var(\epsilon|\mathbf{x}) = \pi^2/3$ in the ordered *logit* model and $Var(\epsilon|\mathbf{x}) = 1$ in the ordered *probit* model. While these assumption identify the variance, the mean of $y^\star$ is still unidentified. We can assume either $\tau_1 = 0$ or $\alpha = 0$. Understanding the different parameterizations is important since different software uses different parameterizations.

2.3.2 ML estimation

The probability of whatever value of y is, therefore, in general case

$$p_n = \begin{cases} Pr(y_n = 1|\mathbf{x}_n, \beta, \tau) & \text{if } y = 1 \\ \vdots & \vdots \\ Pr(y_n = m|\mathbf{x}_n, \beta, \tau) & \text{if } y = m \\ \vdots & \vdots \\ Pr(y_n = J|\mathbf{x}_n, \beta, \tau) & \text{if } y = J \end{cases}$$

If the observations are independent, the likelihood equation is

$$L(\beta, \tau|\mathbf{y}, \mathbf{X}) = \prod_{n=1}^{N} p_n \ .$$

Taking logs, the log likelihood is

$$lnL(\beta, \tau|\mathbf{y}, \mathbf{X}) = \sum_{j=1}^{J} \sum_{y_n=j} ln(F(\tau_J - x_n'\beta) - F(\tau_{J-1} - x_n'\beta)) \ .$$

This equation can be maximized with numerical methods to estimate the τ's and the β's.

It is very important to know which parametrization the software package you use is applied. Programs such as LIMDEP assume that $\tau_1 = 0$ and estimate β_0 while programs such as SAS's LOGISTIC and STATA assume that $\beta_0 = 0$ and estimate τ_1. The choice of parameterization does not affect estimates of the slopes, but does affect the estimates of β_0 and the τ's.

2.3.3 Interpretation

Similar to Binary Regression Model, there is no single approach to interpretation of the Ordered Regression Model. You should search for an elegant and concise way to summarize your results.

- *Predicted probabilities*

  The predicted probability that $y = m$ given $\mathbf{x}$ is

  $$\widehat{Pr}(y = m|\mathbf{x}) = F(\tau_m - \mathbf{x}'\beta) - F(\tau_{m-1} - \mathbf{x}'\beta)) \ .$$

  These probabilities can be used in a variety of ways to show the relationships between the independent

variables and the dependent categories.

- 1) *Determining the mean and range of predicted probabilities*

It is useful to begin by examining the mean, minimum a and maximum predicted probabilities over the sample

$$\begin{aligned}
\text{mean } \widehat{Pr}(y = m|\mathbf{x}) &= \frac{1}{N}\sum_{n=1}^{N} \widehat{Pr}(y = m|\mathbf{x}_n) \\
\min \widehat{Pr}(y = m|\mathbf{x}) &= \min_n \widehat{Pr}(y = m|\mathbf{x}_n) \\
\max \widehat{Pr}(y = m|\mathbf{x}) &= \max_n \widehat{Pr}(y = m|\mathbf{x}_n)
\end{aligned}$$

- 2) *Plotting predicted probabilities*

  Similar to Binary Regression Model, with a single independent variable the entire probability curve can be plotted. When there are more variables, the effect of a single variables can be examined while the remaining variables are held constant. Figure 13 shows an example.

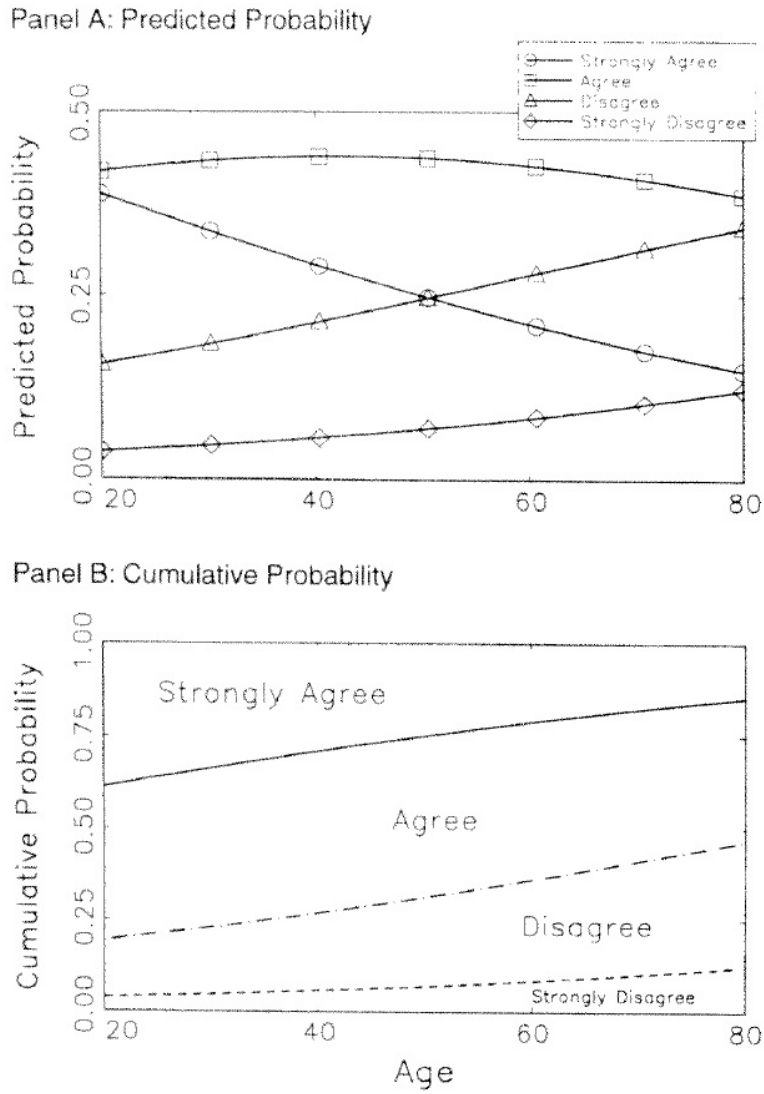
- 3) *Plotting cumulative probabilities*

The cumulative probability is the probability that the outcome is less than or equal to some value:

$$Pr(y \leq m|\mathbf{x}) = \sum_{j=1}^{m} Pr(y = j|\mathbf{x}) = F(\tau_m - \mathbf{x}'\beta)$$

Figure 13 shows an example. Note that the cumulative probabilities in this example "stack" the corresponding probabilities from the top panel, and, show the overall increase with age in negative attitudes toward the statement.

Figure 13: Predicted and cumulative probabilities



Source: Long (1997)

These predicted and cumulative probabilities can be presented for different categories (men and women) in similar plots to those presented in Figure 13 or in a table.

- *Partial change in predicted probabilities*

The partial change or marginal effect is defined in the Ordered Regression Model as

$$\begin{aligned}\frac{\partial Pr(y=m|\mathbf{x})}{\partial x_k} &= \frac{\partial F(\tau_m - \mathbf{x}'\beta)}{\partial x_k} - \frac{\partial F(\tau_{m-1} - \mathbf{x}'\beta)}{\partial x_k} \\ &= \beta_k[f(\tau_{m-1} - \mathbf{x}'\beta) - f(\tau_m - \mathbf{x}'\beta)] \ .\end{aligned}$$

This marginal effect is the slope of the curve relating x_k to $Pr(y = m|\mathbf{x})$, holding all other variables constant. The sign of the marginal effect is not necessarily the same as the sign of β, since $f(\tau_{m-1} - \mathbf{x}'\beta) - f(\tau_m - \mathbf{x}'\beta)$ can be negative. The marginal effect of x_k can change sign as x_k changes. This is seen in Figure 13 for the probability of agreeing. Initially, the slope of the probability curve is positive, indicating positive marginal effect of age. Around age 40, the marginal turns negative, showing that increasing age decreases the probability of agreeing.

Since the marginal effect depends on the levels of all variables, we must decide on which values of the variables to use when computing the effect. We have similar alternatives presented in previous section for Binary Regression Model. One method is to compute the average over all observations:

$$\text{mean } \frac{\partial Pr(y = m|\mathbf{x})}{\partial x_k} = \frac{1}{N}\sum_{n=1}^{N} \beta_k[f(\tau_{m-1} - \mathbf{x}_n'\beta) - f(\tau_m - \mathbf{x}_n'\beta)] \ .$$

More commonly, the marginal effect is computed at the mean values of all variables:

$$\frac{\partial Pr(y = m|\bar{\mathbf{x}})}{\partial x_k} = \beta_k[f(\tau_{m-1} - \bar{\mathbf{x}}'\beta) - f(\tau_m - \bar{\mathbf{x}}'\beta)] \ .$$

In general, the marginal effect **does not** indicate the change in the probability that would be observed for a unit change in x_k. If you conclude for example:

For women each additional 10 years of age increases the probability of disagreeing with a specific statement by 0.32,

then the value 0.32 is 10 times the marginal effect of age for the given outcome. Beware that this interpretation of the marginal effect is only possible when the probability curve is nearly linear.

- *Discrete change*

Interpretation using marginal effects can be misleading when probability curve is changing rapidly or when an independent variable is a dummy variable. For the Ordered Regression Model the measures of discrete change are much more informative.

Discrete change is defined as

$$\frac{\triangle Pr(y = m|\bar{\mathbf{x}})}{\triangle x_k} = Pr(y \leq m|\bar{\mathbf{x}}, x_k = x_E) - Pr(y \leq m|\bar{\mathbf{x}}, x_k = x_S)$$

The change is interpreted as:

When x_k changes from x_S to x_E, the predicted probability of outcome m changes by $\triangle Pr(y = m|\mathbf{x})/\triangle x_k$, holding all other variables constant at $\mathbf{x}$. Since the model is nonlinear, the value of the discrete change depends on three factors:

- the level of all of the variables that are not changing,
- the value at which x_k starts,
- the amount of change in x_k.

Most frequently, each continuous variable except x_k is held at its mean. For dummy independent variables, the change might be computed for both values of the variable.

The effect of a variable can be summarized by computing the average of the absolute values of the changes across all of the outcome categories. The absolute value is taken since the sum of the changes without taking the absolute value is necessarily zero. The *average absolute discrete change* is

$$\bar{\triangle} = \frac{1}{J} \sum_{j=1}^{J} \left| \frac{\triangle Pr(y = j|\bar{\mathbf{x}})}{\triangle x_k} \cdot \right|$$

This indicator can be used for comparison of effects on attitudes towards specific question.

2.3.4 Example: Ordered logit - working mothers and childern

```
##############################################################################
#                                                                            #
# Ordered logit                                                              #
#                                                                            #
# This example is based on a question from the 1977 and 1989 General Social  #
# Survey. Respondents were asked to evaluate the following statement:        #
# "A working mother can establish just as warm and secure of a               #
# relationship with her child as a mother who does not work".                #
# Responses were coded as:                                                   #
#   1=Strongly Disagree (SD)                                                 #
#   2=Disagree (D)                                                           #
#   3=Agree (A)                                                              #
#   4=Strongly Agree (SA)                                                    #

# Preliminaries
rm(list = ls(all = TRUE))             # Clear workspace
Sys.setenv(LANG = "en")               # Set language
setwd(".")                            # Set working directory
#----------------------------------------------------------------------------#
# Loading Libraries                                                          #
#----------------------------------------------------------------------------#
# Read Data Stored by 'Minitab', 'S', 'SAS', 'SPSS', 'Stata', ...
# OK read.dta() included in this library
if (!require("foreign")) {install.packages("foreign")}; library(foreign)

# Testing Linear Regression Models
# waldtest() included in this library
if (!require("lmtest")) {install.packages("lmtest")}; library(lmtest)

# Harrell Miscellaneous - Contains many functions useful for data analysis
if (!require("Hmisc")) {install.packages("Hmisc")}; library(Hmisc)

# Analysis of Overdispersed Data
if (!require("aod")) {install.packages("aod")}; library(aod)

# A Package for Estimation of Ordered Generalized Linear Models.
#if (!require("oglmx")) {install.packages("oglmx")}; library(oglmx)

# Support Functions and Datasets for Venables and Ripley's MASS
# polr() included in this library
if (!require("MASS")) {install.packages("MASS")}; library(MASS)
```

```
Loading required package: MASS
```

```
# Miscellaneous Functions for "Grid" Graphics
# grid.arrange() included in this library
if (!require("gridExtra")) {install.packages("gridExtra")}; library(gridExtra)
```

```
Loading required package: gridExtra
```

```
if (!require("tidyverse")) {install.packages("tidyverse")}; library(tidyverse)
```

```
Loading required package: tidyverse
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v lubridate 1.9.4     v tibble    3.3.0
v purrr     1.1.0     v tidyr     1.3.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::combine()   masks gridExtra::combine()
x dplyr::filter()    masks stats::filter()
x dplyr::lag()       masks stats::lag()
x dplyr::select()    masks MASS::select()
x dplyr::src()       masks Hmisc::src()
x dplyr::summarize() masks Hmisc::summarize()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
#-------------------------------------------------------------------------------#


odwarm       <- read.dta("https://raw.githubusercontent.com/discretechoice/Data/refs/heads/main/ordwarm
odwarm$yr89  <- as.numeric(odwarm$yr89)-1
odwarm$male  <- as.numeric(odwarm$male)-1
odwarm$white <- as.numeric(odwarm$white)-1
summary(odwarm)
```

```
  warm          yr89             male            white             age
 1SD:297   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :18.00
 2D :723   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:1.0000   1st Qu.:31.00
 3A :856   Median :0.0000   Median :0.0000   Median :1.0000   Median :42.00
 4SA:417   Mean   :0.3986   Mean   :0.4649   Mean   :0.8766   Mean   :44.94
           3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:58.00
           Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :89.00
       ed             prst         warmlt2     warmlt3       warmlt4
 Min.   : 0.00   Min.   :12.00   D,A,SA:1996   A,SA:1273   SA    : 417
 1st Qu.:11.00   1st Qu.:30.00   SD    : 297   D,SD:1020   A,D,SD:1876
 Median :12.00   Median :37.00
 Mean   :12.22   Mean   :39.59
 3rd Qu.:14.00   3rd Qu.:50.00
 Max.   :20.00   Max.   :82.00
```

```r
#  variable name   type   format      label      variable label                          #
#  ---------------------------------------------------------------------------------------#
#  warm            byte   %10.0g      sd2sa      Mom can have warm relations with child  #
#  yr89            byte   %10.0g      yrlbl      Survey year: 1=1989 0=1977              #
#  male            byte   %10.0g      sexlbl     Gender: 1=male 0=female                 #
#  white           byte   %10.0g      race2lbl   Race: 1=white 0=not white               #
#  age             byte   %10.0g                 Age in years                            #
#  ed              byte   %10.0g                 Years of education                      #
#  prst            byte   %10.0g                 Occupational prestige                   #

# Variable |       Obs        Mean    Std. Dev.       Min        Max      #
# -------------+--------------------------------------------------------  #
# warm |      2293    2.607501    .9282156          1          4          #
# yr89 |      2293    .3986044    .4897178          0          1          #
# male |      2293    .4648932    .4988748          0          1          #
# white|      2293    .8765809    .3289894          0          1          #
# age  |      2293    44.93546    16.77903         18         89          #
```

```r
# -------------+-------------------------------------------------------- #
# ed    |      2293    12.21805    3.160827          0          20        #
# prst  |      2293    39.58526    14.49226         12          82        #


addmargins(table(odwarm$warm))
```

```
 1SD   2D   3A  4SA  Sum
 297  723  856  417 2293
```

```r
warm.ologit <- polr(warm ~ yr89 + male + white + age + ed + prst,
                    data = odwarm)
summary(warm.ologit)
```

```
Re-fitting to get Hessian

Call:
polr(formula = warm ~ yr89 + male + white + age + ed + prst,
    data = odwarm)

Coefficients:
          Value Std. Error t value
yr89   0.523912   0.079899   6.557
male  -0.733309   0.078483  -9.344
white -0.391140   0.118381  -3.304
age   -0.021666   0.002469  -8.777
ed     0.067176   0.015975   4.205
prst   0.006072   0.003293   1.844

Intercepts:
       Value    Std. Error t value
1SD|2D  -2.4654   0.2389   -10.3188
2D|3A   -0.6309   0.2333    -2.7042
3A|4SA   1.2618   0.2340     5.3919

Residual Deviance: 5689.825
AIC: 5707.825
```

```r
# If the dependent variable does not vary within one of the categories #
# of an independent variable, there will be a problem with estimation. #
# To see what happens, let's transform the prestige variable prst      #
# into a dummy variable                                                #

odwarm$dumprst    <- as.numeric(odwarm$prst < 20 & odwarm$warm == "1SD")
addmargins(table(odwarm$dumprst,odwarm$warm))
```

```
       1SD   2D   3A  4SA  Sum
  0    257  723  856  417 2253
  1     40    0    0    0   40
  Sum  297  723  856  417 2293
```

```r
warm.ologit2      <- polr(warm ~ yr89 + male + white + age + ed + dumprst,
                          data = odwarm)
```

```r
summary(warm.ologit2)
```

```
Re-fitting to get Hessian

Call:
polr(formula = warm ~ yr89 + male + white + age + ed + dumprst,
    data = odwarm)

Coefficients:
            Value Std. Error    t value
yr89      0.52687  8.060e-02  6.537e+00
male     -0.72518  7.929e-02 -9.146e+00
white    -0.42407  1.197e-01 -3.542e+00
age      -0.02106  2.446e-03 -8.608e+00
ed        0.07214  1.331e-02  5.419e+00
dumprst -15.42528  1.593e-07 -9.683e+07

Intercepts:
       Value         Std. Error    t value
1SD|2D -2.776200e+00  2.436000e-01 -1.139710e+01
2D|3A  -8.423000e-01  2.364000e-01 -3.563200e+00
3A|4SA  1.061500e+00  2.366000e-01  4.487100e+00

Residual Deviance: 5544.524
AIC: 5562.524
```

```r
# Therefore, it is useful to check the dummy variables
addmargins(table(odwarm$yr89 , odwarm$warm))
```

```
       1SD   2D   3A  4SA  Sum
  0    236  460  466  217 1379
  1     61  263  390  200  914
  Sum  297  723  856  417 2293
```

```r
addmargins(table(odwarm$male , odwarm$warm))
```

```
       1SD   2D   3A  4SA  Sum
  0    139  323  461  304 1227
  1    158  400  395  113 1066
  Sum  297  723  856  417 2293
```

```r
addmargins(table(odwarm$white, odwarm$warm))
```

```
       1SD   2D   3A  4SA  Sum
  0     25   82  107   69  283
  1    272  641  749  348 2010
  Sum  297  723  856  417 2293
```

```r
# Testing                                                                    #

warm.ologit <- polr(warm ~ yr89 + male + white + age + ed + prst,
                    data = odwarm)
summary(warm.ologit)
```

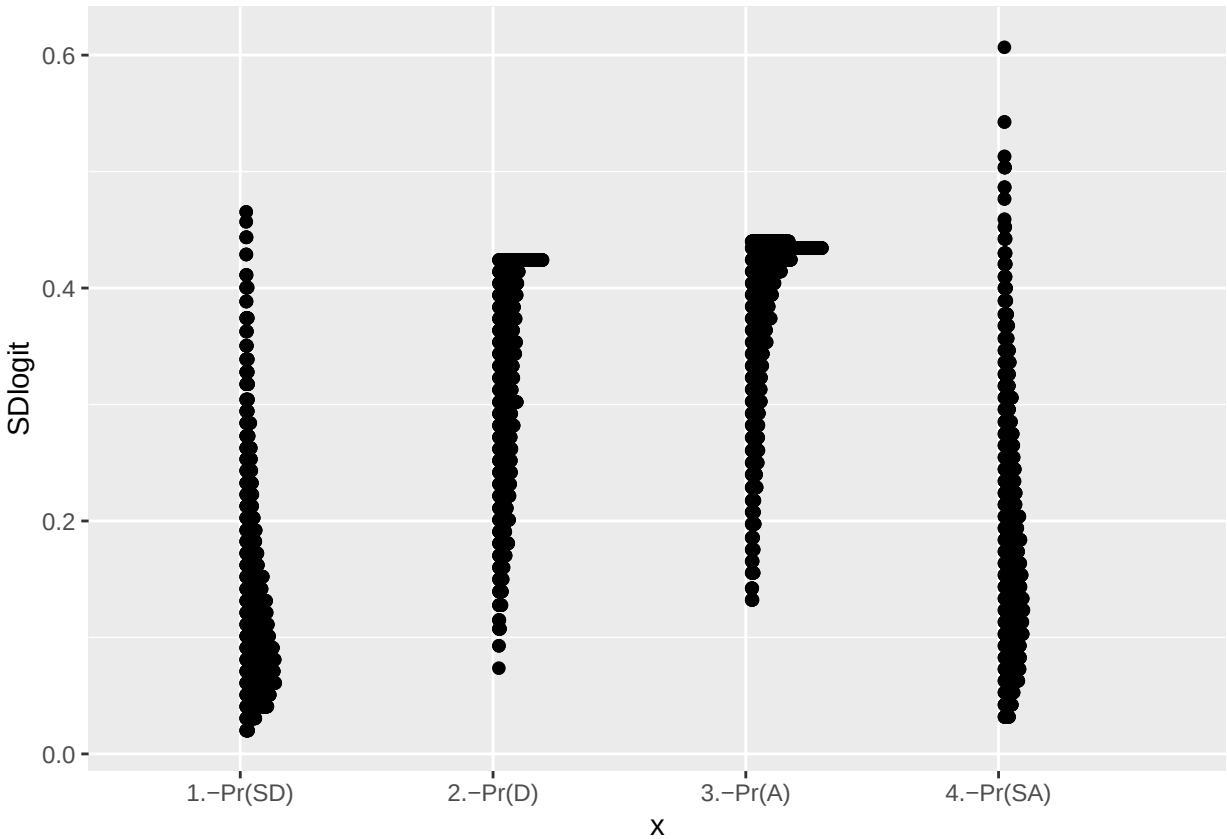
```
Re-fitting to get Hessian

Call:
polr(formula = warm ~ yr89 + male + white + age + ed + prst,
    data = odwarm)

Coefficients:
          Value Std. Error t value
yr89   0.523912   0.079899   6.557
male  -0.733309   0.078483  -9.344
white -0.391140   0.118381  -3.304
age   -0.021666   0.002469  -8.777
ed     0.067176   0.015975   4.205
prst   0.006072   0.003293   1.844

Intercepts:
       Value    Std. Error t value
1SD|2D  -2.4654   0.2389   -10.3188
2D|3A   -0.6309   0.2333    -2.7042
3A|4SA   1.2618   0.2340     5.3919

Residual Deviance: 5689.825
AIC: 5707.825
```

```r
# Gender significantly affects attitudes toward working mothers
# (z = -9.34, p < 0.01 for a two-tailed test).

waldtest(warm.ologit, c("age","white","male"), test = "Chisq")
```

```
Re-fitting to get Hessian

Wald test

Model 1: warm ~ yr89 + male + white + age + ed + prst
Model 2: warm ~ yr89 + ed + prst
  Res.Df Df Chisq Pr(>Chisq)
1   2284
2   2287 -3 166.6  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# The hypothesis that the demographic effects of age, race, and
# gender are simultaneously equal to zero can be rejected at
# the .01 level (W = 166.62, df = 3, p < .01).

# Interpretation                                                          #

# Examining predicted probabilities within the sample provides
# a first, quick check of the model
prediction <- predict(warm.ologit, type = "probs")
SDlogit    <- prediction[,1]
Dlogit     <- prediction[,2]
Alogit     <- prediction[,3]
SAlogit    <- prediction[,4]
```

```r
ggplot(odwarm) +
  geom_dotplot(aes(x = "1.-Pr(SD)", y = SDlogit) , binaxis = "y", stackdir = "up", stackratio = 0.015, 
  geom_dotplot(aes(x = "2.-Pr(D)" , y = Dlogit)  , binaxis = "y", stackdir = "up", stackratio = 0.015, 
  geom_dotplot(aes(x = "3.-Pr(A)" , y = Alogit)  , binaxis = "y", stackdir = "up", stackratio = 0.015, 
  geom_dotplot(aes(x = "4.-Pr(SA)", y = SAlogit) , binaxis = "y", stackdir = "up", stackratio = 0.015, 
```




```r
# We might want to examine the predicted probabilities #
# for individuals with the following characteristics:  #
# Working class men in 1977 who are near retirement.   #
newdata <- data.frame(
  yr89   = 0 ,
  male   = 1 ,
  white  = 1 ,
  age    = 64,
  ed     = 16,
  prst   = 20
)
predict(warm.ologit, newdata, type = "probs")
```

```
       1SD         2D         3A        4SA
0.24038862 0.42421594 0.26473970 0.07065573
```

```r
# Young, highly educated women with prestigious jobs.
newdata2 <- data.frame(
  yr89  = 1 ,
  male  = 0 ,
  white = 1 ,
```

```r
  age   = 30,
  ed    = 24,
  prst  = 80
)
predict(warm.ologit, newdata2, type = "probs")
```

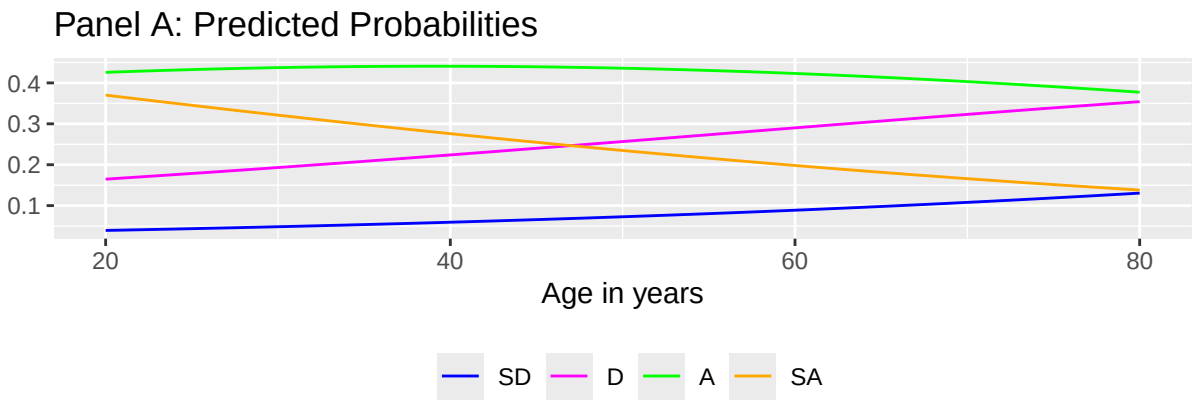
```
       1SD         2D         3A        4SA
0.01718953 0.08151664 0.32223625 0.57905758
```

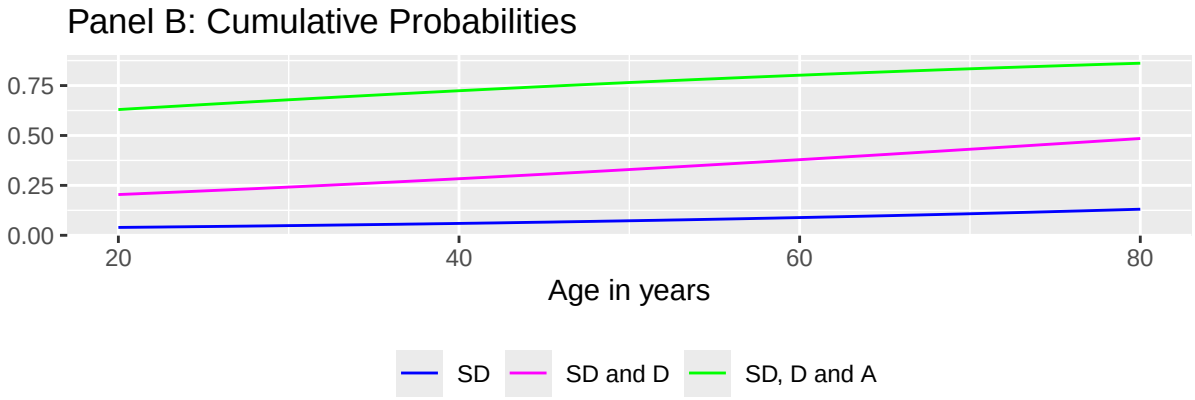
```r
#                                  #
# Graphing predicted probabilities #
#                                  #
# Getting the predicions           #
# Individuals analysed             #
newdata3 <- data.frame(
  yr89  = 1                         ,
  male  = 0                         ,
  white = 1                         ,
  age   = 20                        ,
  ed    = quantile(odwarm$ed, 0.5)  ,
  prst  = quantile(odwarm$prst, 0.5)
)
# Age range
ages   <- seq(20, 80, by = 0.5)
probSA <- rep(0, length(ages))
probA  <- rep(0, length(ages))
probD  <- rep(0, length(ages))
probSD <- rep(0, length(ages))
for (i in 1:length(ages)) {
  newdata3$age <- ages[i]
  prediction   <- predict(warm.ologit, newdata3, type = "probs")
  probSD[i]    <- prediction[1]
  probD[i]     <- prediction[2]
  probA[i]     <- prediction[3]
  probSA[i]    <- prediction[4]
}
# We store the information in a dataframe to be used with ggplot
pred.probs <- as.data.frame(cbind(ages, probSA, probA, probD, probSD),
                            colnames=c("ages", "probSA", "probA", "probD", "probSD"))
head(pred.probs)
```

```
  ages    probSA     probA     probD     probSD
1 20.0 0.3701374 0.4258092 0.1647211 0.03933230
2 20.5 0.3676154 0.4265661 0.1660748 0.03974367
3 21.0 0.3651007 0.4273045 0.1674356 0.04015917
4 21.5 0.3625933 0.4280243 0.1688036 0.04057882
5 22.0 0.3600933 0.4287253 0.1701787 0.04100268
6 22.5 0.3576009 0.4294074 0.1715609 0.04143077
```

```r
# Creating the plots
panelA <- ggplot(pred.probs)                                          +
  geom_line(aes(ages, probSD, colour = "SD"))                         +
  geom_line(aes(ages, probD , colour = "D" ))                         +
  geom_line(aes(ages, probA , colour = "A" ))                         +
  geom_line(aes(ages, probSA, colour = "SA"))                         +
```

```r
  scale_colour_manual("", breaks = c("SD","D","A","SA"),
                      values = c("blue", "magenta", "green", "orange")) +
  theme(legend.position = "bottom")                                      +
  xlab("Age in years") + ylab("")                                        +
  labs(title = "Panel A: Predicted Probabilities")

panelB <- ggplot(pred.probs)                                            +
  geom_line(aes(ages, probSD              , colour = "SD"           )) +
  geom_line(aes(ages, probD+probSD        , colour = "SD and D"     )) +
  geom_line(aes(ages, probA+probD+probSD  , colour = "SD, D and A" )) +
  scale_colour_manual("", breaks = c("SD", "SD and D", "SD, D and A"),
                      values = c("blue", "magenta", "green"))         +
  theme(legend.position = "bottom")                                   +
  xlab("Age in years") + ylab("")                                     +
  labs(title = "Panel B: Cumulative Probabilities")
# Showing the result
grid.arrange(panelA, panelB, nrow = 2)
```






```r
# Panel A plots the predicted probabilities and shows that with age the      #
# probability of SA decreases rapidly while the probability of D (and to     #
# a lesser degree SD) increases. Panel B plots the cumulative probabilities. #
# Since both panels present the same information, which one you use          #
# is largely a matter of personal preference                                 #


#-------------------------------------------------------------------------------#
## Marginal Effects at Representative Values (MER)                             #
```

```r
#-------------------------------------------------------------------------------#

# 1. Calculate medians for continuous variables (excluding NA values)
median_age <- median(odwarm$age, na.rm = TRUE)
median_ed <- median(odwarm$ed, na.rm = TRUE)
median_prst <- median(odwarm$prst, na.rm = TRUE)

# 2. Define the specific point for Marginal Effects (MER)
# yr89=1, male=0, white=1, age=median, ed=median, prst=median
at_values <- data.frame(
  yr89 = 1,
  male = 0,
  white = 1,
  age = median_age,
  ed = median_ed,
  prst = median_prst
)

# 3. Compute the Marginal Effects at Representative Values (MER)
level_labels <- c("1SD", "2D", "3A", "4SA")
mer_output <- list()

for (i in 1:length(level_labels)) {

  # Compute ME for the specific level at the defined 'at_values' point
  mer_at_level <- margins(
    warm.ologit,
    at = at_values,
    type = "response",
    vce = "delta",
    level = level_labels[i]
  )

  # Summarize the results and CORRECT THE COLUMN NAMES
  # Use 'factor' which is the common column name for the variable in 'margins' summary
  mer_sum_level <- summary(mer_at_level) %>%
    as_tibble() %>%
    # **FIX: Use 'factor' instead of 'term'**
    select(factor, AME, SE, z, p) %>%
    rename(`term` = factor, `Marg. Eff` = AME, `Std. error` = SE,
           `t value` = z, `Pr(>|t|)` = p) %>%
    # Reorder the columns and ensure p-value formatting
    select(term, `Marg. Eff`, `Std. error`, `t value`, `Pr(>|t|)`) %>%
    mutate(`Pr(>|t|)` = format.pval(`Pr(>|t|)`, digits = 3))

  mer_output[[i]] <- mer_sum_level
}
```

```
Re-fitting to get Hessian


Re-fitting to get Hessian
```

```
Re-fitting to get Hessian
```

```
Re-fitting to get Hessian
```

```r
# 4. Print the results in the desired format
cat("\n##################################################################################\n")
```

```
##################################################################################
```

```r
cat("## Marginal Effects at Representative Values (MER) for warm.ologit\n")
```

```
## Marginal Effects at Representative Values (MER) for warm.ologit
```

```r
cat(paste("## Computed at: yr89=1, male=0, white=1, age=", round(median_age, 2),
          ", ed=", round(median_ed, 2), ", prst=", round(median_prst, 2), "\n", sep=""))
```

```
## Computed at: yr89=1, male=0, white=1, age=42, ed=12, prst=37
```

```r
cat("##################################################################################\n")
```

```
##################################################################################
```

```r
cat("Marginal Effects on Pr(Outcome==1)\n")
```

```
Marginal Effects on Pr(Outcome==1)
```

```r
print(mer_output[[1]], row.names = FALSE)
```

```
# A tibble: 6 x 5
  term  `Marg. Eff` `Std. error` `t value` `Pr(>|t|)`
  <chr>       <dbl>        <dbl>     <dbl> <chr>
1 age      0.00126      0.000355      3.55 0.000
2 ed      -0.00390      0.000775     -5.03 0
3 male     0.0426       0.00972       4.38 0
4 prst    -0.000352     0.000209     -1.69 0.091
5 white    0.0227       0.00965       2.35 0.019
6 yr89    -0.0304       0.00680      -4.47 0
```

```r
cat("-----------------------------------\n")
```

```
-----------------------------------
```

```r
cat("Marginal Effects on Pr(Outcome==2)\n")
```

```
Marginal Effects on Pr(Outcome==2)
```

```r
print(mer_output[[2]], row.names = FALSE)
```

```
# A tibble: 6 x 5
  term  `Marg. Eff` `Std. error` `t value` `Pr(>|t|)`
  <chr>       <dbl>        <dbl>     <dbl> <chr>
1 age      0.00126      0.000355      3.55 0.000
2 ed      -0.00390      0.000775     -5.03 0
3 male     0.0426       0.00972       4.38 0
4 prst    -0.000352     0.000209     -1.69 0.091
5 white    0.0227       0.00965       2.35 0.019
6 yr89    -0.0304       0.00680      -4.47 0
```

```r
cat("-------------------------------------\n")
```

```
-------------------------------------
```

```r
cat("Marginal Effects on Pr(Outcome==3)\n")
```

```
Marginal Effects on Pr(Outcome==3)
```

```r
print(mer_output[[3]], row.names = FALSE)
```

```
# A tibble: 6 x 5
  term  `Marg. Eff` `Std. error` `t value` `Pr(>|t|)`
  <chr>       <dbl>        <dbl>     <dbl> <chr>
1 age      0.00126      0.000355      3.55 0.000
2 ed      -0.00390      0.000775     -5.03 0
3 male     0.0426       0.00972       4.38 0
4 prst    -0.000352     0.000209     -1.69 0.091
5 white    0.0227       0.00965       2.35 0.019
6 yr89    -0.0304       0.00680      -4.47 0
```

```r
cat("-------------------------------------\n")
```

```
-------------------------------------
```

```r
cat("Marginal Effects on Pr(Outcome==4)\n")
```

```
Marginal Effects on Pr(Outcome==4)
```

```r
print(mer_output[[4]], row.names = FALSE)
```

```
# A tibble: 6 x 5
  term  `Marg. Eff` `Std. error` `t value` `Pr(>|t|)`
  <chr>       <dbl>        <dbl>     <dbl> <chr>
1 age      0.00126      0.000355      3.55 0.000
2 ed      -0.00390      0.000775     -5.03 0
3 male     0.0426       0.00972       4.38 0
4 prst    -0.000352     0.000209     -1.69 0.091
5 white    0.0227       0.00965       2.35 0.019
6 yr89    -0.0304       0.00680      -4.47 0
```

```r
cat("-------------------------------------\n")
```

```
-------------------------------------
```

2.4 Nominal Outcomes: Multinomial Logit

When a variable is nominal, the categories cannot be ordered. Models for nominal outcomes are often used when the dependent variable is ordinal. If a dependent variable is ordinal and a model for nominal variables is used, there is a loss of efficiency since information is being ignored. On the other hand, when a method for ordinal variables is applied to a nominal dependent variable, the resulting estimates are biased or even nonsensical. If there is any question about the ordinality of the dependent variable, the potential loss of efficiency in using models for nominal outcomes is outweighted by avoiding potential bias.

The multinomial logit model (MNLM) can be thought of as simultaneously estimating binary logits for all possible comparisons among the outcome categories. With three outcomes, MNLM is roughly equivalent to running three binary logits comparing outcomes 1 to 2, 1 to 3, and 2 to 3.

With four outcomes you get 1-2, 1-3, 1-4, 2-3, 2-4, 3-4. The number of comparisons can be overwhelming and interpretation difficult when the number of possible outcomes increases.

Consider a nominal outcome y with categories A, B, and C, with N_A, N_B and N_C observations in each category. Assume that there is a single independent variable x. To examine the effect of x on the odds of A versus B, we select $N_A + N_B$ observations with outcomes A and B and estimate binary logit:

$$ln\left[\frac{Pr(A|x)}{Pr(B|x)}\right] = \beta_{0,A|B} + \beta_{1,A|B}x \ .$$

According to this equation a unit increase in x changes the odds of A versus B by a factor of $exp(\beta_{1,A|B})$. The remaining comparisons can be analyzed in the same way.

$$\begin{aligned} ln\left[\frac{Pr(B|x)}{Pr(C|x)}\right] &= \beta_{0,B|C} + \beta_{1,B|C}x \\ ln\left[\frac{Pr(A|x)}{Pr(C|x)}\right] &= \beta_{0,A|C} + \beta_{1,A|C}x \ . \end{aligned}$$

Given that

$$\begin{aligned} ln\left[\frac{Pr(A|x)}{Pr(C|x)}\right] &= ln\left[\frac{Pr(A|x)}{Pr(B|x)} \cdot \frac{Pr(B|x)}{Pr(C|x)}\right] \\ &= ln\left[\frac{Pr(A|x)}{Pr(B|x)}\right] + ln\left[\frac{Pr(B|x)}{Pr(C|x)}\right] \ , \end{aligned}$$

we get

$$(\beta_{0,A|B} + \beta_{1,A|B}x) + (\beta_{0,B|C} + \beta_{1,B|C}x) = (\beta_{0,A|C} + \beta_{1,A|C}x) \ .$$

Or, looking at the intercepts and slopes separately,

$$\beta_{0,A|B} + \beta_{0,B|C} = \beta_{0,A|C}$$
$$\beta_{1,A|B} + \beta_{1,B|C} = \beta_{1,A|C} \ .$$

Therefore, some of the comparisons are redundant.These equalities describe necessary relationships among the parameters in the population. They will not hold with a sample estimates from three binary logits because they are based on different samples. Therefore, in the MNLM, all of the logits are estimated simultaneously, which enforces the logical relationship among the parameters and uses the data more efficiently.

2.4.1 The MNLM as a probability model

Let y be the dependent variable with J nominal outcomes. Let $Pr(y = m|\mathbf{x})$ be the probability of observing outcome m given $\mathbf{x}$.

- Assume that $Pr(y = m|\mathbf{x})$ is a function of the linear combination $\mathbf{x}'\beta_m$. The vector $\beta_m = (\beta_{0m} \cdots \beta_{km} \cdots \beta_{Km})'$ includes the intercept β_{0m} and coefficients β_{km} for the effect of x_k on outcome m.
- To ensure that the probabilities are nonnegative, we take the exponential of $\mathbf{x}'\beta_m$. While the result is nonnegative, the sum $\sum_{j=1}^{J} exp(\mathbf{x}'\beta_m)$ does not equal 1 which it must for all probabilities.
- In order to make the probabilities sum to 1, we divide $exp(\mathbf{x}'\beta_m)$ by $\sum_{j=1}^{J} exp(\mathbf{x}'\beta_j)$:

$$Pr(y_n = m|\mathbf{x}_n) = \frac{exp(\mathbf{x}_n'\beta_m)}{\sum_{j=1}^{J} exp(\mathbf{x}_n'\beta_j)} \ .$$

- With this normalization, it follows that $\sum_{m=1}^{J} Pr(y = m|\mathbf{x}_n) = 1$.

While the probabilities now sum to 1, they are unidentified since more than one set of parameters generates the same probabilities of the observed outcomes. To identify the model, we must impose constraints on the β's. We can assume sum over alternatives is zero, $\sum_{j=1}^{J} \beta_j = 0$ or more commonly, one of the β's is zero, such as $\beta_1 = 0$. In this case the model is commonly written as

$$Pr(y_n = 1|\mathbf{x}_n) = \frac{1}{1 + \sum_{j=2}^{J} exp(\mathbf{x}_n'\beta_j)} \tag{3}$$

$$Pr(y_n = m|\mathbf{x}_n) = \frac{exp(\mathbf{x}_n'\beta_m)}{1 + \sum_{j=2}^{J} exp(\mathbf{x}_n'\beta_j)} \text{ for } m > 1 \ . \tag{4}$$

2.4.2 The MNLM as a Discrete Choice Model

The discrete choice model is based on the principle that an individual chooses the outcome that maximizes the utility gained from that choice. The utility derived from choice j for individual n equals

$$U_{nj} = V_{nj} + \epsilon_{nj} \ ,$$

where V_{nj} is the average utility associated with choice j for individual n and ϵ_{nj} is the random error term associated with that choice. Then, if there are only two alternatives,

$$\begin{aligned} Pr(y_n = 1) &= Pr(U_{n1} > U_{n2}) \\ &= Pr(V_{n1} + \epsilon_{n1} > V_{n2} + \epsilon_{n2}) \\ &= Pr(\epsilon_{n2} - \epsilon_{n1} < V_{n1} - V_{n2}) \end{aligned}$$

The specific form of the discrete choice model is determined by the assumed distribution of ϵ and the specification of how V_m is related to measured variables. We can get the MNLM if

$$U_{nj} = V_{nj} + \epsilon_{nj} \tag{5}$$

$$= \mathbf{x}_n' \beta_j + \epsilon_{nj} \tag{6}$$

and ϵ's are independent and have a type I extreme-value distribution:

$$f(\epsilon) = exp[-\epsilon - exp(-\epsilon)] \ .$$

Under these assumption probabilities $Pr(y_n = 1|\mathbf{x})$ are those defined in (3) and (4), that is

$$Pr(y_n = 1|\mathbf{x}_n) = \frac{1}{1 + \sum_{j=2}^{J} exp(\mathbf{x}_n' \beta_j)} \tag{7}$$

$$Pr(y_n = m|\mathbf{x}_n) = \frac{exp(\mathbf{x}_n' \beta_m)}{1 + \sum_{j=2}^{J} exp(\mathbf{x}_n' \beta_j)} \text{ for } m > 1 \ . \tag{8}$$

2.4.3 Example: Nominal outcome - occupation and education

```
#  EXAMPLE:                          #
#  Nominal outcome: MNLM             #
#  educ:      Years of education     #
#  occ3:      1 - Manual             #
#             2 - White Collar       #
#             3 - Proffesional       #
```

$$
\begin{aligned}
U_{n,man} &= ASC_{man} + \beta_{man} educ_n + \varepsilon_{n,man} \\
U_{n,wc} &= ASC_{wc} + \beta_{wc} educ_n + \varepsilon_{n,wc} \\
U_{n,prof} &= ASC_{prof} + \beta_{prof} educ_n + \varepsilon_{n,prof}
\end{aligned} \tag{9}
$$

Identification of the model requires that one of the β's be set to zero (benchmark category) and the variance of the errors be normalized to a specific value.

The model can be estimated using the `mlogit` function from the `mlogit` package. The key to understanding the output is the line with `reflevel = "Manual"` that is the benchmark group. This means that $ASC_{man} = 0$ and $\beta_{man} = 0$ and the utilities for the other two categories are compared to the `Manual` category:

$$
\begin{aligned}
U_{n,man} &= 0 + 0\, educ_n + \varepsilon_{n,man} \\
U_{n,wc} &= ASC_{wc} + \beta_{wc} educ_n + \varepsilon_{n,wc} \\
U_{n,prof} &= ASC_{prof} + \beta_{prof} educ_n + \varepsilon_{n,prof}
\end{aligned}
$$

That is why the probabilites are calculated as:

$$
\begin{aligned}
Pr(y_n = man | educ_n) &= \frac{exp(0)}{exp(0) + exp(ASC_{wc} + \beta_{wc} educ_n) + exp(ASC_{prof} + \beta_{prof} educ_n)} \\
&= \frac{1}{1 + exp(ASC_{wc} + \beta_{wc} educ_n) + exp(ASC_{prof} + \beta_{prof} educ_n)} \\
Pr(y_n = wc | educ_n) &= \frac{exp(ASC_{wc} + \beta_{wc} educ_n)}{exp(0) + exp(ASC_{wc} + \beta_{wc} educ_n) + exp(ASC_{prof} + \beta_{prof} educ_n)} \\
&= \frac{exp(ASC_{wc} + \beta_{wc} educ_n)}{1 + exp(ASC_{wc} + \beta_{wc} educ_n) + exp(ASC_{prof} + \beta_{prof} educ_n)} \\
Pr(y_n = prof | educ_n) &= \frac{exp(ASC_{prof} + \beta_{prof} educ_n)}{exp(0) + exp(ASC_{wc} + \beta_{wc} educ_n) + exp(ASC_{prof} + \beta_{prof} educ_n)} \\
&= \frac{exp(ASC_{prof} + \beta_{prof} educ_n)}{1 + exp(ASC_{wc} + \beta_{wc} educ_n) + exp(ASC_{prof} + \beta_{prof} educ_n)}
\end{aligned}
$$

```r
# Preliminaries
rm(list = ls(all = TRUE))                                   # Clear workspace
Sys.setenv(LANG = "en")                                     # Set language

#-------------------------------------------------------------------------------#
# Example: Type of occupation and education
#-------------------------------------------------------------------------------#
#
# Preliminaries
rm(list = ls(all = TRUE))             # Clear workspace
Sys.setenv(LANG = "en")               # Set language
setwd(".")
#-------------------------------------------------------------------------------#
# Loading Libraries                                                             #
#-------------------------------------------------------------------------------#
# Read Data Stored by 'Minitab', 'S', 'SAS', 'SPSS', 'Stata', ...
# read.dta() included in this library
if (!require("foreign")) {install.packages("foreign")}; library(foreign)

# Testing Linear Regression Models
# waldtest() included in this library
if (!require("lmtest")) {install.packages("lmtest")}; library(lmtest)

# Multinomial Logit Models
if (!require("mlogit")) {install.packages("mlogit")}; library(mlogit)
```

```
Loading required package: mlogit

Loading required package: dfidx
```

```r
# Multinomial Logit Models, with or without Random Effects or Overdispersion
# mclogit() included in this library
if (!require("mclogit")) {install.packages("mclogit")}; library(mclogit)
```

```
Loading required package: mclogit

Loading required package: Matrix


Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```r
#  EXAMPLE:                            #
#  Nominal outcome: MNLM               #
#  ed:        Years of education       #
#  occ3:      1 - Manual               #
#             2 - White Collar         #
#             3 - Proffesional         #

data <- read.dta("https://raw.githubusercontent.com/discretechoice/Data/refs/heads/main/nomintro2.dta")
summary(data)
```

```
       ed              occ3          wc_man      prof_man       prof_wc
 Min.   : 3.00   Manual  :184   Manual  :184   Manual:184   WhiteCol: 41
```

```
 1st Qu.:12.00   WhiteCol: 41   WhiteCol: 41   Prof  :112   Prof    :112
 Median :12.00   Prof    :112   NA's    :112   NA's  : 41   NA's    :184
 Mean   :13.09
 3rd Qu.:15.00
 Max.   :20.00
```

```r
# ---------------------------------------------------------------------------#
# storage  display     value                                                 #
# variable name   type   format      label      variable label               #
# ---------------------------------------------------------------------------#
# ed              byte   %10.0g                 Years of education           #
# occ3            byte   %9.0g       locc3                                   #
# wc_man          byte   %9.0g       lwm                                     #
# prof_man        byte   %9.0g       lpm                                     #
# prof_wc         byte   %9.0g       lpw                                     #
# ---------------------------------------------------------------------------#

table(data$ed, data$occ3)
```

```
     Manual WhiteCol Prof
  3       1        0    0
  6       8        0    0
  7       3        0    1
  8      10        1    1
  9       9        0    0
  10      9        1    0
  11     15        2    2
  12     77       19   13
  13     16        4   10
  14     23        1   14
  15      6        5    7
  16      7        6   32
  17      0        1    6
  18      0        1    8
  19      0        0   13
  20      0        0    5
```

```r
# Multinomial logit                                          #
datat       <- mlogit.data(data   = data   ,
                           choice = "occ3",
                           shape  = "wide" )
mlogit      <- mlogit(occ3 ~ 1 |  ed    ,
                      data     = datat ,
                      reflevel ="Manual")
summary(mlogit)
```

```
Call:
mlogit(formula = occ3 ~ 1 | ed, data = datat, reflevel = "Manual",
    method = "nr")

Frequencies of alternatives:choice
  Manual     Prof WhiteCol
 0.54599  0.33234  0.12166
```

```
nr method
5 iterations, 0h:0m:0s
g'(-H)^-1g = 0.000191
successive function values within tolerance limits

Coefficients :
                       Estimate Std. Error z-value  Pr(>|z|)
(Intercept):Prof     -10.211192   1.106961 -9.2245 < 2.2e-16 ***
(Intercept):WhiteCol  -5.232601   1.096113 -4.7738 1.808e-06 ***
ed:Prof                0.719566   0.080515  8.9370 < 2.2e-16 ***
ed:WhiteCol            0.300073   0.084138  3.5664 0.0003619 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -248.15
McFadden R^2:  0.22717
Likelihood ratio test : chisq = 145.89 (p.value = < 2.22e-16)
```

```
#  The key to understanding the output is the line with                    #
#  reflevel = "Manual" that is the comparison group.                       #
```

2.4.4 ML estimation and testing

Regardless of which approach is used to derive the model, the equation for the probabilities of the MNLM (3) and (4) are the same. Let p_n be the probability of observing whatever value of y was actually observed for the ith observation. If the observations are independent, the likelihood equation is

$$L(\beta_2, \beta_3, \ldots, \beta_J | \mathbf{y}, \mathbf{X}) = \prod_{n=1}^{N} p_n \ .$$

Taking into account the definition of probabilities p_n, the β's are introduced into the likelihood equation as follows

$$L(\beta_2, \beta_3, \ldots, \beta_J | \mathbf{y}, \mathbf{X}) = \prod_{m=1}^{J} \prod_{y_n=m} \frac{exp(\mathbf{x}_n' \beta_m)}{\sum_{j=1}^{J} exp(\mathbf{x}_n' \beta_j)} \ ,$$

where $\prod_{y_n=m}$ is the product over all cases for which y_n is equal to m. Taking logs, we obtain log-likelihood equation which can be maximized with numerical methods to estimate the β's.

There are two tests that are useful when using the MNLM. The first is a **test that the effect of a variable is** 0. With J dependent categories, there are $J-1$ parameters $\beta_{k,m|r}$ associated with each variable x_k. The hypothesis that x_k does not affect the dependent variable can be written as

$$H_0 : \beta_{k,1|r} = \cdots = \beta_{k,J|r} = 0 \ .$$

Since $\beta_{k,r|r} = 0$ is necessarily 0, the hypothesis imposes constraints on $J-1$ parameters.

The second is a **test of whether a pair of outcome categories can be combined**. If none of the x_k's significantly affects the odds of outcome m versus outcome n, we say that m and n are indistinguishable with respect to the variables in the model. If $\beta_{1,m|n}, \ldots, \beta_{K,m|n}$ are the coefficients for x_1 through x_K from the logit of the alternative m versus n, then the hypothesis that outcomes m and n are indistinguishable corresponds to

$$H_0 : \beta_{1,m|n} = \cdots = \beta_{K,m|n} = 0 \ .$$

These hypotheses can be tested with either a Wald, LR or LM test.

2.4.5 Example: Nominal outcome - occupation, education, experience and race

$$\begin{aligned}
U_{n,man} &= ASC_{man} + \beta_{1,man} white_n + \beta_{2,man} ed_n + \beta_{3,man} exper_n + \varepsilon_{n,man} \\
U_{n,bluec} &= ASC_{bluec} + \beta_{1,bluec} white_n + \beta_2 ed_n + \beta_{3,bluec} exper_n + \varepsilon_{n,bluec} \\
U_{n,craft} &= ASC_{craft} + \beta_{1,craft} white_n + \beta_{2,craft} ed_n + \beta_{3,craft} exper_n + \varepsilon_{n,craft} \\
U_{n,whitec} &= ASC_{whitec} + \beta_{1,whitec} white_n + \beta_{2,whitec} ed_n + \beta_{3,whitec} exper_n + \varepsilon_{n,whitec} \\
U_{n,prof} &= ASC_{prof} + \beta_{1,prof} white_n + \beta_{2,prof} ed_n + \beta_{3,prof} exper_n + \varepsilon_{n,prof}
\end{aligned} \tag{10}$$

Identification is in this case achieved by setting coeffcientes of the `prof` level to zero:

$$\begin{aligned}
U_{n,man} &= ASC_{man} + \beta_{1,man} white_n + \beta_{2,man} ed_n + \beta_{3,man} exper_n + \varepsilon_{n,man} \\
U_{n,bluec} &= ASC_{bluec} + \beta_{1,bluec} white_n + \beta_{2,bluec} ed_n + \beta_{3,bluec} exper_n + \varepsilon_{n,bluec} \\
U_{n,craft} &= ASC_{craft} + \beta_{1,craft} white_n + \beta_{2,craft} ed_n + \beta_{3,craft} exper_n + \varepsilon_{n,craft} \\
U_{n,whitec} &= ASC_{whitec} + \beta_{1,whitec} white_n + \beta_{2,whitec} ed_n + \beta_{3,whitec} exper_n + \varepsilon_{n,whitec} \\
U_{n,prof} &= 0 + 0\, white_n + 0\, ed_n + 0\, exper_n + \varepsilon_{n,prof}
\end{aligned} \tag{11}$$

Test that the effect of `exper` is zero:

$$H_0 : \begin{pmatrix} \beta_{3,\text{man}} \\ \beta_{3,\text{bluec}} \\ \beta_{3,\text{craft}} \\ \beta_{3,\text{whitec}} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Test of whether a pair of outcome categories (`menial` and `craft`) can be combined:

$$H_0 : \begin{pmatrix} \beta_{1,\text{menial}} \\ \beta_{2,\text{menial}} \\ \beta_{3,\text{menial}} \end{pmatrix} = \begin{pmatrix} \beta_{1,\text{bluec}} \\ \beta_{2,\text{bluec}} \\ \beta_{3,\text{bluec}} \end{pmatrix}$$

```r
#-----------------------------------------------------------------------------#
# Example: Nominal outcome - Occupation, education, experience and race
#-----------------------------------------------------------------------------#
#
# Preliminaries
rm(list = ls(all = TRUE))             # Clear workspace
Sys.setenv(LANG = "en")               # Set language
setwd(".")
#-----------------------------------------------------------------------------#
# Loading Libraries                                                           #
#-----------------------------------------------------------------------------#
# Read Data Stored by 'Minitab', 'S', 'SAS', 'SPSS', 'Stata', ...
# read.dta() included in this library
if (!require("foreign")) {install.packages("foreign")}; library(foreign)

# Testing Linear Regression Models
# waldtest() included in this library
if (!require("lmtest")) {install.packages("lmtest")}; library(lmtest)

# Multinomial Logit Models
if (!require("mlogit")) {install.packages("mlogit")}; library(mlogit)

# Multinomial Logit Models, with or without Random Effects or Overdispersion
# mclogit() included in this library
if (!require("mclogit")) {install.packages("mclogit")}; library(mclogit)

# Analysis of Overdispersed Data
if (!require("aod")) {install.packages("aod")}; library(aod)
                                                          #
#                                                                             #
# The 1982 General Social Survey asked respondents their occupation,          #
# which we recoded into five broad categories: menial jobs (M),               #
# blue collar jobs (B), craft jobs (C), white collar jobs (W), and            #
# professional jobs (P). Three independent variables are considered:          #
# white indicating the race of the respondent, ed measuring years of education, #
# and exper measuring years of work experience                                #

data <- read.dta("https://raw.githubusercontent.com/discretechoice/Data/refs/heads/main/nomocc2.dta")
summary(data)
```

```
       occ          white              ed            exper
 Menial  : 31   Min.   :0.0000   Min.   : 3.00   Min.   : 2.0
 BlueCol : 69   1st Qu.:1.0000   1st Qu.:12.00   1st Qu.: 9.0
 Craft   : 84   Median :1.0000   Median :12.00   Median :17.0
```

```
 WhiteCol: 41   Mean   :0.9169   Mean   :13.09   Mean   :20.5
 Prof    :112   3rd Qu.:1.0000   3rd Qu.:15.00   3rd Qu.:31.0
                Max.   :1.0000   Max.   :20.00   Max.   :66.0
```

```r
datat       <- mlogit.data(data=data, choice = "occ", shape = "wide")
mlogit      <- mlogit(occ ~ 1 | white + ed + exper,
                      data = datat, reflevel = "Prof")
summary(mlogit)
```

```
Call:
mlogit(formula = occ ~ 1 | white + ed + exper, data = datat,
    reflevel = "Prof", method = "nr")

Frequencies of alternatives:choice
    Prof  BlueCol    Craft   Menial WhiteCol
0.332344 0.204748 0.249258 0.091988 0.121662

nr method
5 iterations, 0h:0m:0s
g'(-H)^-1g = 3.46E-08
gradient close to zero

Coefficients :
                       Estimate Std. Error z-value  Pr(>|z|)
(Intercept):BlueCol  12.2595643  1.6681445  7.3492 1.994e-13 ***
(Intercept):Craft    10.4269779  1.5179434  6.8691 6.459e-12 ***
(Intercept):Menial   11.5183308  1.8493561  6.2283 4.715e-10 ***
(Intercept):WhiteCol  5.2797225  1.6840058  3.1352  0.001717 **
white:BlueCol        -0.5378027  0.7996033 -0.6726  0.501210
white:Craft          -1.3019629  0.6474160 -2.0110  0.044324 *
white:Menial         -1.7743065  0.7550543 -2.3499  0.018778 *
white:WhiteCol       -0.2029212  0.8693072 -0.2334  0.815429
ed:BlueCol           -0.8782767  0.1005446 -8.7352 < 2.2e-16 ***
ed:Craft             -0.6850365  0.0892996 -7.6712 1.710e-14 ***
ed:Menial            -0.7788519  0.1146293 -6.7945 1.087e-11 ***
ed:WhiteCol          -0.4256943  0.0922192 -4.6161 3.910e-06 ***
exper:BlueCol        -0.0309296  0.0144086 -2.1466  0.031824 *
exper:Craft          -0.0079671  0.0127055 -0.6271  0.530620
exper:Menial         -0.0356509  0.0180370 -1.9765  0.048093 *
exper:WhiteCol       -0.0010550  0.0143582 -0.0735  0.941424
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -426.8
McFadden R^2:  0.16288
Likelihood ratio test : chisq = 166.09 (p.value = < 2.22e-16)
```

```r
# listcoef presents estimates for all combinations of outcome categories. #
# Since listcoef can generate a lot of output,                            #
# you can include a list of variables and only coefficients for those     #
# variables will be listed.                                               #
coef(mlogit)
```

```
 (Intercept):BlueCol    (Intercept):Craft   (Intercept):Menial
```

```
        12.259564315         10.426977945         11.518330753
(Intercept):WhiteCol        white:BlueCol          white:Craft
         5.279722474         -0.537802683         -1.301962900
        white:Menial       white:WhiteCol           ed:BlueCol
        -1.774306494         -0.202921210         -0.878276652
            ed:Craft            ed:Menial          ed:WhiteCol
        -0.685036546         -0.778851948         -0.425694273
       exper:BlueCol          exper:Craft         exper:Menial
        -0.030929648         -0.007967091         -0.035650894
      exper:WhiteCol
        -0.001055038
```

```r
# Testing the effects of the independent variables #
mlogit  <- mlogit(occ ~ 1|  ed + white + exper , data = datat, reflevel="Prof")
mlogit2 <- mlogit(occ ~ 1|  ed         + exper , data = datat, reflevel="Prof")
mlogit3 <- mlogit(occ ~ 1|       white + exper , data = datat, reflevel="Prof")
mlogit4 <- mlogit(occ ~ 1|  ed + white         , data = datat, reflevel="Prof")

# Easiest way
waldtest(mlogit,mlogit2)
```

```
Wald test

Model 1: occ ~ 1 | ed + white + exper
Model 2: occ ~ 1 | ed + exper
  Res.Df Df  Chisq Pr(>Chisq)
1    321
2    325 -4 8.1492    0.08626 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# The effect of race on occupation is significant at the .10 level #
waldtest(mlogit,mlogit3)
```

```
Wald test

Model 1: occ ~ 1 | ed + white + exper
Model 2: occ ~ 1 | white + exper
  Res.Df Df  Chisq Pr(>Chisq)
1    321
2    325 -4 84.968  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# The effect of education is significant at the .01 level          #
waldtest(mlogit,mlogit4)
```

```
Wald test

Model 1: occ ~ 1 | ed + white + exper
Model 2: occ ~ 1 | ed + white
  Res.Df Df  Chisq Pr(>Chisq)
1    321
2    325 -4 7.9949    0.09176 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# The effect of exp on occupation is significant at the .10 level  #


# Wald tests of the null hypothesis that two categories      #
# can be combined for all combinations of outcome categories #

# Occupation  |      Freq.     Percent        Cum. #
# ------------+----------------------------------- #
# Menial      |         31        9.20        9.20 #
# BlueCol     |         69       20.47       29.67 #
# Craft       |         84       24.93       54.60 #
# WhiteCol    |         41       12.17       66.77 #
# Prof        |        112       33.23      100.00 #
# ------------+----------------------------------- #

# Menial can be combined with BlueCol?

# Are these coefficients equal?
c(coef(mlogit)[5], coef(mlogit)[9] , coef(mlogit)[13])
```

```
   ed:BlueCol white:BlueCol exper:BlueCol
  -0.87827665   -0.53780268   -0.03092965
```

```r
c(coef(mlogit)[7], coef(mlogit)[11], coef(mlogit)[15])
```

```
   ed:Menial white:Menial exper:Menial
 -0.77885195  -1.77430649  -0.03565089
```

```r
# l%*%b = c(0,0,0)
l= rbind( c(0,0,0,0,-1,0,1,0,0,0,0,0,0,0,0,0),
          c(0,0,0,0,0,0,0,0,-1,0,1,0,0,0,0,0),
          c(0,0,0,0,0,0,0,0,0,0,0,0,-1,0,1,0))

wald.test(b     = coef(mlogit),
          Sigma = vcov(mlogit),
          L     = l           ,
          H0    = c(0,0,0)    )
```

```
Wald test:
----------

Chi-squared test:
X2 = 4.0, df = 3, P(> X2) = 0.26
```

```r
#  We cannot reject the hypothesis that outcomes Menial and BlueCol #
#  are indistinguishable                                            #
```

2.4.6 Interpretation

In even a simple MNLM, there are a lot of parameters. With three outcomes and five independent variables, there are 12 unique parameters.

- *Predicted probabilities*

The predicted probability that $y = m$ given $\mathbf{x}$ is

$$Pr(y_n = m|\mathbf{x}_n) = \frac{exp(\mathbf{x}_n'\beta_m)}{\sum_{j=1}^{J} exp(\mathbf{x}_n'\beta_j)} \ .$$

where $\beta_1 = 0$. Probabilities can be, therefore, computed at a variety of values and presented in various ways. You can compute the mean, minimum and maximum probabilities over the sample. To examine the effect of a single x_k, you can fix all variables except x_k at some level and plot the predicted probability as x_k changes over its range.

- *Partial change*

For continuous independent variables, the partial change in the probability is

$$\frac{\partial Pr(y = m|\mathbf{x})}{\partial x_k} = Pr(y_n = m|\mathbf{x}) \left[\beta_{km} - \sum_{j=1}^{J} \beta_{kj} Pr(y = j|\mathbf{x}) \right] \ . \tag{12}$$

The value of the marginal effect depends on the values of all independent variables and on the coefficients for each outcome. According to (12) the marginal effect of x_k on m does not need to have the same sign as the corresponding coefficient β_{km}. Further, as x_k changes, the sign of the marginal can change.

- *Discrete change*

Measures of discrete change in probabilities are an effective method of interpretation that can be used for continuous and dummy independent variables. The change in the predicted probability when x_k changes from x_S to x_E is

$$\frac{\triangle Pr(y = m|\mathbf{x})}{\triangle x_k} = Pr(y = m|\mathbf{x}, x_k = x_E) - Pr(y = m|\mathbf{x}, x_k = x_S) \ .$$

For a change in variable x_k from x_S to x_E, the predicted probability of outcome m changes by $\triangle Pr(y = m|\mathbf{x})/\triangle x_k$, holding all other variables constant. Similar to binary and ordered regression models, the amount of discrete change in the probability for a change in x_k depends on

- the amount of change in x_k
- the starting value of x_k
- the values of all other variables.

There can be used many alternatives explained in previous sections.

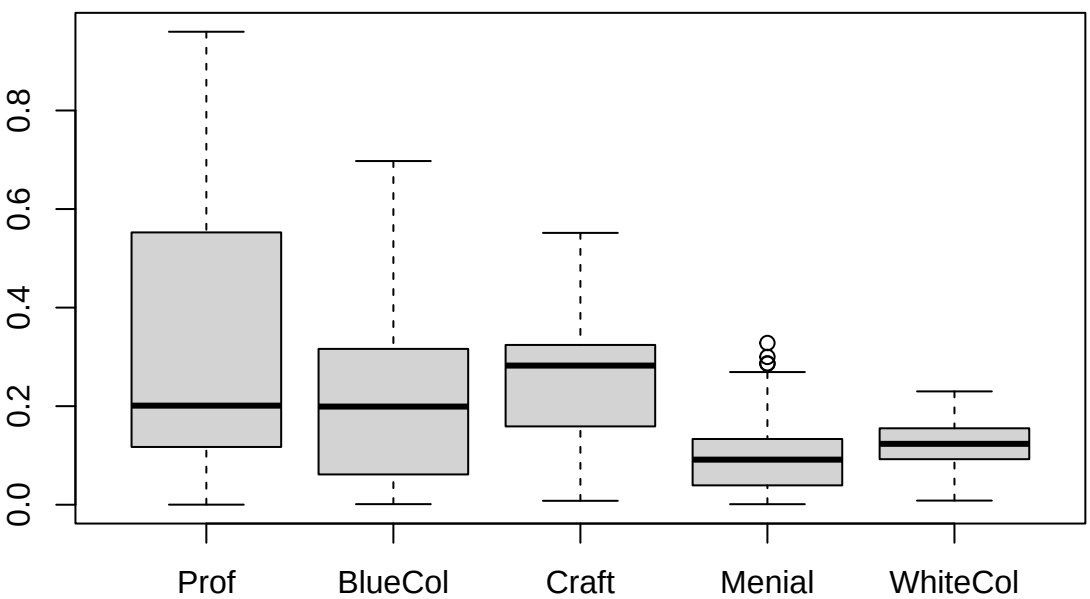
After a decision is made on how to compute the discrete change, there will be J values of change, one for each of the outcomes. These changes can be summarized by computing the *average absolute discrete change*:

$$\bar{\triangle} = \frac{1}{J}\sum_{j=1}^{J}\left|\frac{\triangle Pr(y=j|\bar{\mathbf{x}})}{\triangle x_k}\cdot\right|$$

```
##################
# Interpretation #
##################
# Observed shares of the outcome variable
summary(mlogit)$freq/sum(summary(mlogit)$freq)
```

```
choice
      Prof    BlueCol      Craft     Menial   WhiteCol
0.33234421 0.20474777 0.24925816 0.09198813 0.12166172
```

```
mlogit <- mlogit(occ ~ 1| white + ed + exper,
                 data = datat, reflevel = "Prof", print.level = 0)
# predicted probabilities
prob   <- (mlogit)$probabilities
boxplot(prob)
```




```
# we might compute the probabilities of each occupational  #
# category to compare nonwhites and whites who have median #
# education and experience                                 #

datat       <- mlogit.data(data   = data   ,
                           choice = "occ"  ,
                           shape  = "wide")
mlogit      <- mlogit(occ ~ 1 | white + ed + exper,
                      data     = datat ,
                      reflevel = "Prof")
# summary(mlogit)
```

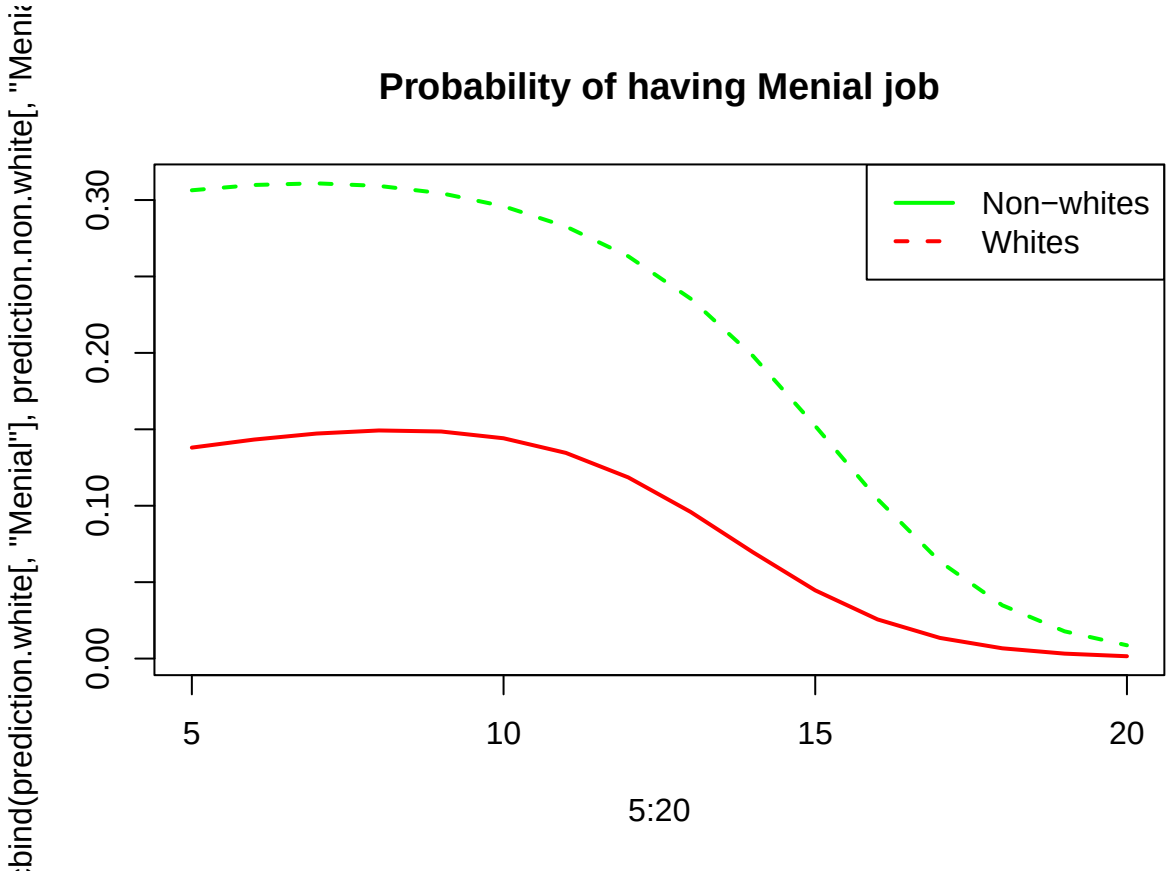
```r
# Prediction for the race: White
# Create a new data set using the first row of the database
newdata.white           <- data[1,]
newdata.white$white[1]  <- 1
newdata.white$ed   [1]  <- quantile(data$ed   ,0.5)[1]
newdata.white$exper[1]  <- quantile(data$exper,0.5)[1]

newdata.white.t    <- mlogit.data(data    = newdata.white,
                                  choice = "occ"        ,
                                  shape  = "wide"       )
prediction.white   <- predict(mlogit                   ,
                              type    = "probs"        ,
                              newdata = newdata.white.t)

# Prediction for the race: non-White                        #
# Create a new data set using the first row of the database #
newdata.non.white           <- data[1,]
newdata.non.white$white[1]  <- 0 # !!!!
newdata.non.white$ed   [1]  <- quantile(data$ed   ,0.5)[1]
newdata.non.white$exper[1]  <- quantile(data$exper,0.5)[1]

newdata.non.white.t    <- mlogit.data(data   = newdata.non.white,
                                      choice = "occ"            ,
                                      shape  = "wide"           )
prediction.non.white   <- predict(mlogit                       ,
                                  type    = "probs"            ,
                                  newdata = newdata.non.white.t)

# Output
output           = matrix(0, nrow = 3, ncol = 5)
rownames(output) = c(
                     "White"     ,
                     "Non-white" ,
                     "Difference"
                    )
colnames(output) = c("Prof", "BlueCol", "Craft", "Menial", "WhiteCol")
output[1,]       = prediction.white
output[2,]       = prediction.non.white
output[3,]       = prediction.white - prediction.non.white

cat("Probabilities for education: "              ,
     as.numeric(quantile(data$ed   ,0.5)[1])     ,
    " and experience: "                          ,
     as.numeric(quantile(data$exper,0.5)[1]),"\n")
```

```
Probabilities for education:  12  and experience:  17
```

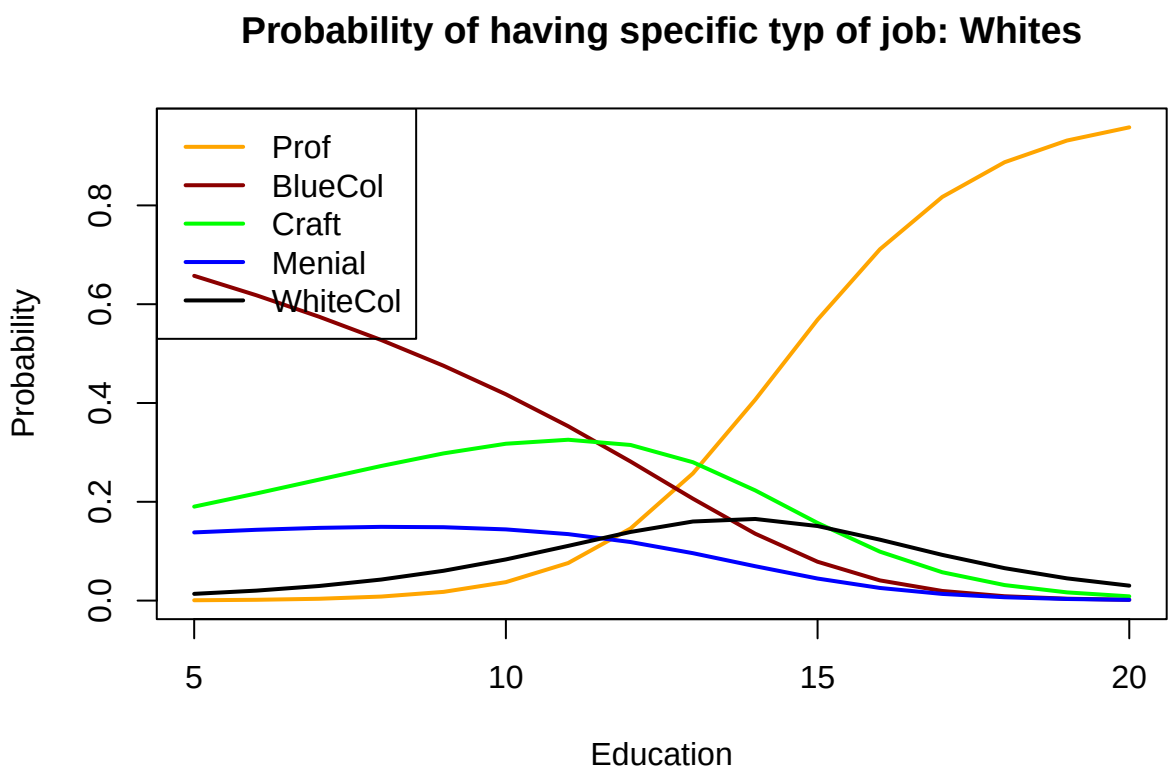
```r
round(output,3)
```

```
            Prof BlueCol  Craft Menial WhiteCol
White      0.146   0.282  0.315  0.119    0.139
Non-white  0.055   0.182  0.436  0.263    0.064
Difference 0.091   0.100 -0.121 -0.145    0.075
```

```r
# We might want to know how white and non-white respondents differ
# in their probability of having a menial job by years of education

# Prediction for the race: White
# Create a new data set using the first row of the database
newdata.white              <- data[1:16,]
newdata.white$white[1:16]  <- rep(1,16)
newdata.white$ed   [1:16]  <- 5:20
newdata.white$exper[1:16]  <- quantile(data$exper,0.5)[1]

newdata.white.t    <- mlogit.data(data   = newdata.white,
                                  choice = "occ"        ,
                                  shape  = "wide"       )
prediction.white   <- predict(mlogit                   ,
                              type    = "probs"        ,
                              newdata = newdata.white.t)

# Prediction fo the race: Non-White                          #
# Create a new data set using the first row of the database #
newdata.non.white              <- data[1:16,]
newdata.non.white$white[1:16]  <- rep(0,16)
newdata.non.white$ed   [1:16]  <- 5:20
newdata.non.white$exper[1:16]  <- quantile(data$exper,0.5)[1]

newdata.non.white.t    <- mlogit.data(data   = newdata.non.white,
                                      choice = "occ"            ,
                                      shape  = "wide"           )
prediction.non.white   <- predict(mlogit                       ,
                                  type    = "probs"            ,
                                  newdata = newdata.non.white.t)

# The columns                                        #
# "prediction.white$menial"                          #
# and                                                #
# "prediction.non.white$menial"                      #
# contain the predicted probabilities                #
# of having menial jobs for nonwhites and whites. #

matplot(y    = cbind(prediction.white[,"Menial"] ,
        prediction.non.white[,"Menial"])         ,
        x    = 5:20                              ,
        type = "l"                               ,
        lty  = 1:2                               ,
        col  = c("red", "green")                 ,
        main = "Probability of having Menial job",
        lwd  = 2)

legend("topright"                        ,
       legend = c("Non-whites","Whites"),
       col    = c("green","red")        ,
       lty    = c(1,2)                  ,
       lwd    = 2 )
```




```
matplot(y    = cbind(prediction.white[,1:5])                    ,
        x    = 5:20                                             ,
        type = "l"                                              ,
        xlab = "Education"                                      ,
        ylab = "Probability"                                    ,
        lty  = rep(1,6)                                         ,
        lwd  = 2                                                ,
        col  = c("orange","dark red","green", "blue","black")   ,
        main = "Probability of having specific typ of job: Whites")

legend("topleft"                                                ,
       legend = c("Prof","BlueCol","Craft","Menial","WhiteCol"),
       col    = c("orange","dark red","green", "blue","black")  ,
       lty    = rep(1,6)                                        ,
       lwd    = 2                                               )
```




```
##############################
# Computing marginal effects #
##############################

datat       <- mlogit.data(data   = data   ,
                           choice = "occ"  ,
                           shape  = "wide")
mlogit      <- mlogit(occ ~ 1 | white + ed + exper,
                      data     = datat ,
                      reflevel = "Prof")
# summary(mlogit)

# Marginal effects at specific data point:                    #
# non-white, median education, median experience              #
# Create a new data set using the first row of the database   #
newdata.white           <- data[1,]
newdata.white$white[1]  <- 0
newdata.white$ed   [1]  <- quantile(data$ed   ,0.5)[1]
newdata.white$exper[1]  <- quantile(data$exper,0.5)[1]

newdata.white.t    <- mlogit.data(data   = newdata.white,
                                  choice = "occ"        ,
                                  shape  = "wide"       )
# marginal effects of "education"
effects(mlogit, covariate= c("ed"   ), data = newdata.white.t)
```

```
        Prof      BlueCol        Craft       Menial     WhiteCol
 0.037964367 -0.034081977  0.002392867 -0.023250133  0.016974876
```

```r
# marginal effects of "white"
effects(mlogit, covariate= c("white"), data = newdata.white.t)
```

```
       Prof     BlueCol       Craft      Menial    WhiteCol
 0.06298164  0.11032176 -0.06822312 -0.16550090  0.06042062
```

```r
# Discrete change of white from 0-1 is very different from marginal change
round(output,3)
```

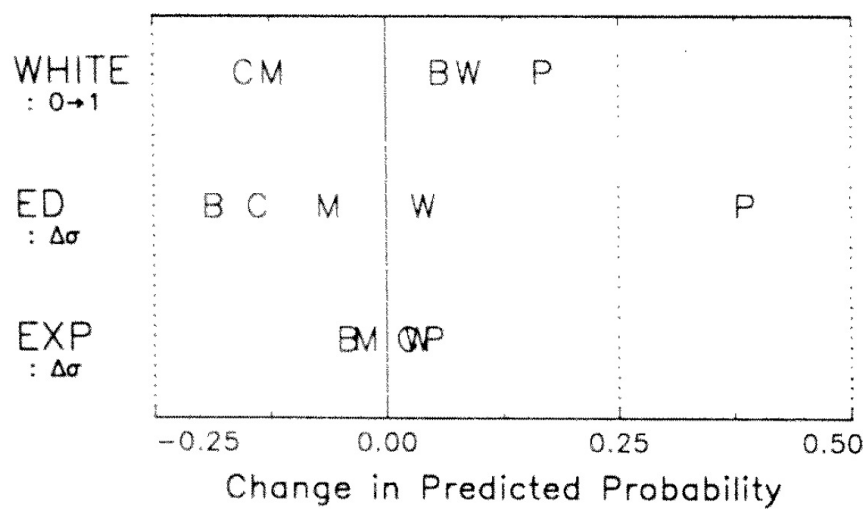
```
            Prof BlueCol  Craft Menial WhiteCol
White      0.146   0.282  0.315  0.119    0.139
Non-white  0.055   0.182  0.436  0.263    0.064
Difference 0.091   0.100 -0.121 -0.145    0.075
```

```r
# marginal effects of "exper"
effects(mlogit, covariate= c("exper"), data = newdata.white.t)
```

```
        Prof      BlueCol        Craft       Menial     WhiteCol
 0.001019379 -0.002248801  0.004612213 -0.004503616  0.001120824
```

Figure 15 shows the change in the probability of an outcome along the horizontal axis, with each independent variable listed on the vertical axis. The letters corresponding to the occupations show the magnitude of the discrete change in the probability of an outcome for a given change in the independent variable, with the other variables held at their means. In this example of type of occupations it is easy to see that the effect of a standard deviation change in education is the largest, with an increase of over 0.35 for professional occupations. The effects of race are also substantial, with average blacks being less likely to enter blue-collar, white-collar, or professional jobs. The expected changes due to a standard deviation change in experience are much smaller and show that experience increases the probabilities of more highly skilled occupations.

Figure 15: Discrete change plot for the MNLM of occupations. Control variables are held at their means, jobs are classified as: M: menial, C: craft, B: blue collar, W: white collar, P: professional



Source: Long (1997)

While the interpretation of the multinomial logit model is complicated when there are more than a few outcome categories, the graphical methods presented above can be used to uncover the complex patterns of effects. In some applications, these plots reveal patterns that can be described easily in a few sentences, and there is little need to present the graphs. In other cases, the patterns are complex, and the plots are valid part of the results.

2.4.7 The conditional logit model

Following table presents an example how the data are constructed for the MNLM. Assume there is a single independent variable z (years of educations of the individual) and three outcomes (occupations: blue collar, white collar and manager). In this case, the data for four individuals might look as follows:

i	Outcome m	Variable z (education)
1	3	$z_1 = 15$
2	1	$z_2 = 12$
3	2	$z_3 = 11$
4	2	$z_4 = 9$

In the MNLM, each explanatory variable has a different effect on each outcome.

The conditional logit model (CLM) is a closely related model in which the coefficients for a variable are the same for each outcome, but the values of the variables differ for each outcome.

An example of how the data are constructed for the CLM is useful for understanding the model. Assume there is a single independent variable z (time of a specific trip) and three outcomes (mode of transport: car, bus and train for a specific trip). For four individuals the data might look as follows:

i	Outcome m	Outcome chosen	Variable z (time)
1	1	0	$z_{11} = 20$
1	2	1	$z_{12} = 70$
1	3	0	$z_{13} = 30$
2	1	1	$z_{21} = 25$
2	2	0	$z_{22} = 75$
2	3	0	$z_{23} = 30$
3	1	1	$z_{31} = 25$
3	2	0	$z_{32} = 77$
3	3	0	$z_{33} = 33$
4	1	0	$z_{41} = 26$
4	2	0	$z_{42} = 78$
4	3	1	$z_{43} = 31$

For each individual, there are three observations corresponding to the three possible outcomes. The differences in the values of z for the different outcomes determine the probability of various choices.

Random Utility Maximization (RUM) models for this type of data can be derived as follows. A decision-maker n out of N individuals faces a choice among J alternatives. We assume that the decision maker n receives a certain level of utility U_{nj} from an alternative j. The decision maker chooses the alternative that provides the greatest utility. Thus, the discrete choice model states that an alternative i is chosen by decision maker n if and only if $U_{ni} > U_{nj}, \forall j \neq i$.

The researcher does not observe the decision maker's utility, but only the final choice, the level of attributes associated with each alternative, and some characteristics of the decision maker, usually socio-demographic characteristics. The utility U_{nj} is then decomposed as

$$U_{nj} = V_{nj} + \varepsilon_{nj} \tag{13}$$

where V_{nj} is often called representative utility, and ε_{nj} is an error term that represents the factors that affect utility U_{nj} but are not included in V_{nj}. The error term ε_{nj} is not known to the researcher and, therefore, it is assumed to be a random term with a joint density of the random vector denoted $f(\varepsilon_n) = f(\varepsilon_{n1}, \varepsilon_{n2}, \ldots \varepsilon_{nJ})$. The probability that decision maker n, in a choice occasion t chooses alternative i is then defined as

$$\begin{aligned} P_{ni} &= Pr\left(U_{ni} > U_{nj} \forall j \neq i\right) \\ &= Pr\left(V_{ni} + \varepsilon_{ni} > V_{nj} + \varepsilon_{nj} \forall j \neq i\right) \\ &= Pr\left(\varepsilon_{nj} - \varepsilon_{ni} < V_{ni} - V_{nj} \forall j \neq i\right) \end{aligned} \tag{14}$$

This is a definition of a cumulative distribution function, specifically, that the probability that the difference of random terms $\varepsilon_{nj} - \varepsilon_{ni}$ is smaller than the difference of representative utilities $V_{ni} - V_{nj}$.

Given the density $f(\varepsilon_n)$ of the error term and defining an indicator function $I(\cdot)$ equal to one when in expression in parentheses is true and zero otherwise, the probability can be rewritten as

$$\begin{aligned} P_{ni} &= Pr\left(\varepsilon_{nj} - \varepsilon_{ni} < V_{ni} - V_{nj} \forall j \neq i\right) \\ &= \int I\left(\varepsilon_{nj} - \varepsilon_{ni} < V_{ni} - V_{nj} \forall j \neq i\right) f\left(\varepsilon_n\right) d\varepsilon_n. \end{aligned} \tag{15}$$

This expression is a multidimensional integral over the density of the random term and it takes different forms for different specifications of $f(\varepsilon_n)$ leading to specific models such as probit, multinomial logit (MNL), nested logit (NL), or mixed logit (MXL). For some specifications (MNL, NL) this integral has a closed form which greatly simplifies its estimation, whereas for some specifications (probit, MXL) the integral does not have a closed form and it must be evaluated numerically by simulation.

Given the expression (15), the identification of a RUM model is achieved in different ways depending on the specific definition of each model, but the underlying restrictions can be summarized in two statements: *the scale of utility is arbitrary* and *only differences in utility matter* (Train (2009, p. p.19)).

Normalization of the model is usually accomplished by normalizing the variance of the error terms. For example, if the error terms are assumed to be independently and identically (*i.i.d.*) distributed with the same variance, then the normalization is simple because the error variance is usually set to a convenient value.

In the case of *i.i.d.* type I extreme value (with location parameter zero and scale one) errors (also called Gumbel distribution), leading to widely used standard MNL, the scale parameter λ is set to be inversely proportional to the common standard deviation σ_ε of the type I extreme value error terms ε_{njt} as it is usually fixed to $\lambda = \pi/\left(\sqrt{6}\sigma_\varepsilon\right)$. Using that normalization, due to the identification, we get $\mathrm{Var}\left(\varepsilon_{njt}\right) = \pi^2/6$. In the case of *i.i.d.* normal errors, leading to independent Probit model, the scale parameter λ is set to $1/\sigma_\varepsilon$ so that $\mathrm{Var}\left(\varepsilon_{njt}\right) = 1$.

Let us now focus on the second, *only differences in utility matter* proposition. According to (14), the absolute level of utility is irrelevant to the choice, since the chosen alternative is determined by the differences between them. That is, if a constant is added to the utility of all alternatives, the alternative with the highest utility will be the same. This implies that the only parameters that can be identified and estimated are those that capture the differences between alternatives.

Let us illustrate this point with an example. Assuming linearity in the attributes, three alternatives $j = 1, 2, 3$, and two attributes ($attr1$ and $attr2$), the model becomes

$$\begin{aligned} U_{n1} &= ASC_1 + \beta_1 attr1_{n1} + \beta_2 attr2_{n1} + \varepsilon_{n1} \\ U_{n2} &= ASC_2 + \beta_1 attr1_{n2} + \beta_2 attr2_{n2} + \varepsilon_{n2} \\ U_{n3} &= \qquad\quad \beta_1 attr1_{n3} + \beta_2 attr2_{n3} + \varepsilon_{n3} \end{aligned} \tag{16}$$

for $n = 1, 2, \ldots, N, t = 1, 2, \ldots, T$. The alternative-specific constants ASC_1 and ASC_2 capture the average effect on utility of all factors not included in the model. We cannot include alternative-specific constants for all alternatives because only differences in utility matter, and therefore only differences in the alternative-specific constants are relevant, not their absolute values.

If ASC_3 were included in the U_{n3} equation, the differences between $U_{n1} - U_{n3}$ and $U_{n2} - U_{n3}$ would be

$$
\begin{aligned}
U_{n1} - U_{n3} = (ASC_1 - ASC_3) &+ \beta_1 \left(attr1_{n1} - attr1_{n3}\right) \\
&+ \beta_2 \left(attr2_{n1} - attr2_{n3}\right) \\
&+ \left(\varepsilon_{n1} - \varepsilon_{n3}\right) \\
U_{n2} - U_{n3} = (ASC_2 - ASC_3) &+ \beta_1 \left(attr1_{n2} - attr1_{n3}\right) \\
&+ \beta_2 \left(attr2_{n2} - attr2_{n3}\right) \\
&+ \left(\varepsilon_{n2} - \varepsilon_{n3}\right).
\end{aligned} \tag{17}
$$

More details in Gujarati (2006), Long (1997), Long and Freese (2001) and Wooldridge (2006).

2.4.8 Example: Conditional logit - transportation mode

The deterministic utility for each transport mode can be written as

$$
\begin{aligned}
V_{n,\text{train}} &= ASC_{\text{train}} + \beta_{\text{time}}\, time_{n,\text{train}} + \beta_{\text{cost}}\, cost_{n,\text{train}}, \\
V_{n,\text{bus}} &= ASC_{\text{bus}} + \beta_{\text{time}}\, time_{n,\text{bus}} + \beta_{\text{cost}}\, cost_{n,\text{bus}}, \\
V_{n,\text{car}} &= \qquad\qquad\quad \beta_{\text{time}}\, time_{n,\text{car}} + \beta_{\text{cost}}\, cost_{n,\text{car}}.
\end{aligned}
$$

and the full random utility is $U_{ni} = V_{ni} + \varepsilon_{ni}$.

Suppose that `time` is in-vehicle travel time and `cost` is the monetary cost of the trip.
In this setting, both variables usually generate disutility, so we typically expect

$$
\beta_{\text{time}} < 0 \qquad \text{and} \qquad \beta_{\text{cost}} < 0.
$$

The marginal rate of substitution (MRS) between time and cost is obtained by asking: How much can cost increase if travel time decreases, while keeping deterministic utility constant?

Using the total differential of utility,

$$
dV = \beta_{\text{time}}\, d(time) + \beta_{\text{cost}}\, d(cost).
$$

Holding utility constant implies $dV = 0$, so $\beta_{\text{time}}\, d(time) + \beta_{\text{cost}}\, d(cost) = 0$. Solving for the trade-off between cost and time gives

$$d(cost) = -\frac{\beta_{\text{time}}}{\beta_{\text{cost}}}\, d(time).$$

Now consider a one-unit **reduction** in travel time, so that $d(time) = -1$. Then the compensating change in cost is

$$d(cost) = -\frac{\beta_{\text{time}}}{\beta_{\text{cost}}}(-1) = \frac{\beta_{\text{time}}}{\beta_{\text{cost}}}.$$

Because both coefficients are typically negative, this quantity is positive.

Therefore, $\beta_{\text{time}}/\beta_{\text{cost}}$ represents the amount of additional cost an individual is willing to accept in exchange for a one-unit reduction in travel time.

More generally, for any non-monetary attribute x, the **willingness to pay for a one-unit increase in that attribute** is

$$WTP_x = -\frac{\beta_x}{\beta_{\text{cost}}}.$$

This formula gives the monetary value of a marginal change in attribute x. Its sign depends on the sign of β_x:

- if $\beta_x > 0$, individuals are willing to pay for an increase in x;
- if $\beta_x < 0$, individuals would need to be compensated for an increase in x, or equivalently they are willing to pay for a decrease in x.

```
################################################################################
#                                                                              #
# Conditional logit                                                            #
#                                                                              #
# In the multinomial logit model, we estimate how                              #
# individual-specific variables  affect the likelihood                         #
# of observing a given outcome. The conditional logit model                    #
# (CLM) allows us to estimate how nominal outcomes                             #
# are affected by characteristics of the outcomes                              #
# that vary across individuals                                                 #

#------------------------------------------------------------------------------#
# Example: Nominal outcome - conditional logit - transportation mode
#------------------------------------------------------------------------------#
#
# Preliminaries
rm(list = ls(all = TRUE))              # Clear workspace
Sys.setenv(LANG = "en")                # Set language
setwd(".")
```

```r
#-----------------------------------------------------------------------------#
# Loading Libraries                                                           #
#-----------------------------------------------------------------------------#
# Read Data Stored by 'Minitab', 'S', 'SAS', 'SPSS', 'Stata', ...
# read.dta() included in this library
if (!require("foreign")) {install.packages("foreign")}; library(foreign)

# Testing Linear Regression Models
# waldtest() included in this library
if (!require("lmtest")) {install.packages("lmtest")}; library(lmtest)

# Multinomial Logit Models
if (!require("mlogit")) {install.packages("mlogit")}; library(mlogit)

# Multinomial Logit Models, with or without Random Effects or Overdispersion
# mclogit() included in this library
if (!require("mclogit")) {install.packages("mclogit")}; library(mclogit)

# Analysis of Overdispersed Data
if (!require("aod")) {install.packages("aod")}; library(aod)
travel <- read.dta("https://raw.githubusercontent.com/discretechoice/Data/refs/heads/main/travel2.dta")
summary(travel)
```

```
       id            mode         train             bus
 Min.   :  1.00   Train:152   Min.   :0.0000   Min.   :0.0000
 1st Qu.: 58.75   Bus  :152   1st Qu.:0.0000   1st Qu.:0.0000
 Median :101.50   Car  :152   Median :0.0000   Median :0.0000
 Mean   :103.89               Mean   :0.3333   Mean   :0.3333
 3rd Qu.:152.50               3rd Qu.:1.0000   3rd Qu.:1.0000
 Max.   :210.00               Max.   :1.0000   Max.   :1.0000
      car              time             invc            choice
 Min.   :0.0000   Min.   : 180.0   Min.   :  2.00   Min.   :0.0000
 1st Qu.:0.0000   1st Qu.: 393.0   1st Qu.: 17.00   1st Qu.:0.0000
 Median :0.0000   Median : 634.0   Median : 30.00   Median :0.0000
 Mean   :0.3333   Mean   : 632.1   Mean   : 33.95   Mean   :0.3333
 3rd Qu.:1.0000   3rd Qu.: 894.0   3rd Qu.: 45.00   3rd Qu.:1.0000
 Max.   :1.0000   Max.   :1440.0   Max.   :109.00   Max.   :1.0000
      ttme            invt              gc             hinc
 Min.   : 0.00   Min.   : 180.0   Min.   : 30.0   Min.   : 2.00
 1st Qu.: 0.00   1st Qu.: 356.5   1st Qu.: 68.0   1st Qu.:15.00
 Median :34.00   Median : 600.0   Median :105.0   Median :30.00
 Mean   :25.25   Mean   : 606.9   Mean   :113.5   Mean   :31.81
 3rd Qu.:44.00   3rd Qu.: 874.0   3rd Qu.:152.0   3rd Qu.:45.00
 Max.   :99.00   Max.   :1440.0   Max.   :269.0   Max.   :72.00
     psize
 Min.   :1.000
 1st Qu.:1.000
 Median :1.000
 Mean   :1.809
 3rd Qu.:2.000
 Max.   :6.000
```

```r
head(travel)
```

```
  id  mode train bus car time invc choice ttme invt gc hinc psize
```

```
1  1 Train     1   0   0  406   31      0   34  372 71   35     1
2  1   Bus     0   1   0  452   25      0   35  417 70   35     1
3  1   Car     0   0   1  180   10      1    0  180 30   35     1
4  2 Train     1   0   0  398   31      0   44  354 84   30     2
5  2   Bus     0   1   0  452   25      0   53  399 85   30     2
6  2   Car     0   0   1  255   11      1    0  255 50   30     2
```

```r
# The dataset **travel2.dta** is a well-known example used in transportation and
# econometrics textbooks. It contains data about how people choose between different
# travel modes - Train, Bus, or Car - for a given trip.
# Each individual appears three times in the data, once for each alternative.
#
# The variables in the dataset are as follows:
#
# * **id** - A number identifying each traveler.
# * **mode** - The travel mode alternative (Train, Bus, or Car).
# * **train** - A dummy variable equal to 1 if the mode is Train, and 0 otherwise.
# * **bus** - A dummy variable equal to 1 if the mode is Bus, and 0 otherwise.
# * **car** - A dummy variable equal to 1 if the mode is Car, and 0 otherwise.
# * **time** - The total travel time for that mode (in minutes).
# * **invc** - The in-vehicle travel cost for that mode (in money units such as dollars or pounds).
# * **choice** - A binary variable equal to 1 if the person actually chose that mode, and 0 otherwise.
# * **ttme** - The time spent outside the vehicle (for example, access, waiting, or transfer time) in m
# * **invt** - The time spent inside the vehicle in minutes.
# * **gc** - The "generalized cost" of the trip, which combines time and money into one measure.
# * **hinc** - Household income of the traveler, usually in thousands of currency units per year.
# * **psize** - Party size, meaning the number of people traveling together.
#
# For each traveler (id), there are three rows - one for each possible travel mode.
# The variables like time, invc, ttme, and invt vary by mode,
# while hinc and psize are personal characteristics that stay the same
# across the three modes.

# We have data on 152 groups of people travelling for        #
# their vacation, choosing between three modes of travel:   #
# train, bus or car.                                        #
# The group is indicated by the variable id.                #
# For each group of travellers, there are three rows of data #
# corresponding to the three choices faced by each group.   #
# Accordingly, we have N * J = 152 * 3 = 456 observations.  #

choice.clogit <- mclogit(cbind(choice,id) ~ train + bus + time + invc,
                         data = travel)
```

```
Iteration 1 - deviance = 183.4829 - criterion = 0.1535652
Iteration 2 - deviance = 163.9999 - criterion = 0.1187267
Iteration 3 - deviance = 161.9664 - criterion = 0.01254748
Iteration 4 - deviance = 161.9223 - criterion = 0.000272104
Iteration 5 - deviance = 161.9223 - criterion = 1.794654e-07
Iteration 6 - deviance = 161.9223 - criterion = 8.437632e-14
converged
```

```r
summary(choice.clogit)
```

```
Call:
mclogit(formula = cbind(choice, id) ~ train + bus + time + invc,
    data = travel)

       Estimate Std. Error z value Pr(>|z|)
train  2.671238   0.453161   5.895 3.75e-09 ***
bus    1.472335   0.400715   3.674 0.000239 ***
time  -0.019145   0.002451  -7.812 5.65e-15 ***
invc  -0.048166   0.011952  -4.030 5.58e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Null Deviance:     334
Residual Deviance: 161.9
Number of Fisher Scoring iterations:  6
Number of observations:  152
```

```r
# The coefficients for time and invc are negative.        #
# This indicates that the longer it takes to travel by a  #
# given mode, the less likely that mode is to be chosen.  #
# Similarly, the more it costs, the less likely a mode is #
# to be chosen.                                           #

choice.clogit.coefs           <- cbind(choice.clogit$coefficients, exp(choice.clogit$coefficients))
colnames(choice.clogit.coefs) <- c("b","e^b")
choice.clogit.coefs
```

```
                b        e^b
train  2.67123795 14.4578563
bus    1.47233479  4.3594015
time  -0.01914530  0.9810368
invc  -0.04816581  0.9529758
```

Car is the base alternative, so its alternative-specific constant is normalized to zero $ASC_{car} = 0$. In the conditional logit model,

$$P_{n,i} = \frac{\exp(V_{n,i})}{\sum_j \exp(V_{n,j})}.$$

Hence, for any two alternatives i and j,

$$\frac{P_{n,i}}{P_{n,j}} = \exp(V_{n,i} - V_{n,j}).$$

This formula is the basis for interpreting both ASCs and slope coefficients.

Interpretation of the ASCs

Consider train versus car:

$$\frac{P_{n,\text{train}}}{P_{n,\text{car}}} = \exp\left(V_{n,\text{train}} - V_{n,\text{car}}\right).$$

Using the utility expressions,

$$V_{n,\text{train}} - V_{n,\text{car}} = ASC_{\text{train}} + \beta_{\text{time}}\left(time_{n,\text{train}} - time_{n,\text{car}}\right) + \beta_{\text{cost}}\left(cost_{n,\text{train}} - cost_{n,\text{car}}\right).$$

If train and car have the same time and cost, then

$$V_{n,\text{train}} - V_{n,\text{car}} = ASC_{\text{train}},$$

so that

$$\frac{P_{n,\text{train}}}{P_{n,\text{car}}} = \exp(ASC_{\text{train}}).$$

Thus, $\exp(ASC_{\text{train}})$ is the odds of choosing train rather than car when observed attributes are equal.

If $ASC_{\text{train}} = 2.671$, then

$$\exp(2.671) \approx 14.46.$$

So, when train and car have the same observed time and cost, the odds of choosing train rather than car are about 14.46 times as large.

Similarly, for bus versus car,

$$\frac{P_{n,\text{bus}}}{P_{n,\text{car}}} = \exp(ASC_{\text{bus}}).$$

If $ASC_{\text{bus}} = 1.472$, then

$$\exp(1.472) \approx 4.36.$$

So, holding observed time and cost equal, the odds of choosing bus rather than car are about 4.36 times as large.

Interpretation of the time coefficient

A slope coefficient gives the **change in the odds** when an attribute changes.

Consider alternative i. Its utility is

$$V_{n,i} = ASC_i + \beta_{\text{time}}\, time_{n,i} + \beta_{\text{cost}}\, cost_{n,i}.$$

Suppose the travel time of alternative i increases by 1 unit, while all other observed attributes remain fixed. Then

$$V'_{n,i} = V_{n,i} + \beta_{\text{time}}.$$

Before the change,

$$\frac{P_{n,i}}{P_{n,j}} = \exp(V_{n,i} - V_{n,j}).$$

After the change,

$$\frac{P'_{n,i}}{P'_{n,j}} = \exp(V'_{n,i} - V_{n,j}).$$

Substituting $V'_{n,i} = V_{n,i} + \beta_{\text{time}}$,

$$\frac{P'_{n,i}}{P'_{n,j}} = \exp(\beta_{\text{time}})\frac{P_{n,i}}{P_{n,j}}.$$

Therefore,

$$\frac{P'_{n,i}/P'_{n,j}}{P_{n,i}/P_{n,j}} = \exp(\beta_{\text{time}}).$$

So, a one-unit increase in the travel time of alternative i multiplies its odds relative to any other alternative by $\exp(\beta_{\text{time}})$.

If $\beta_{\text{time}} = -0.019$, then

$$\exp(-0.019) \approx 0.981.$$

Hence, a one-unit increase in travel time reduces the odds of choosing that alternative by about 1.9%.

Interpretation of the cost coefficient

If $\beta_{\text{cost}} = -0.048$, then

$$\exp(-0.048) \approx 0.953.$$

Hence, a one-unit increase in cost reduces the odds of choosing that alternative by about 4.7%.

```
# For the alternative-specific variables, time and invc,              #
# the odds ratios are the multiplicative effect of a unit             #
# change in a given independent variable on the odds of any           #
# given mode of travel.                                               #
# For example,                                                        #
# Increasing the time of travel by one minute for a given             #
# mode of transportation decreases the odds of using that mode        #
# of travel by a factor of .98 (2%), holding the values for the       #
# other alternatives constant.                                        #
#                                                                     #
# That is, if the time it takes to travel by car increases            #
# by one minute while the time it takes to travel by                  #
# train and bus remain constant, the odds of travelling by             #
# car decrease by 2 percent.                                          #
#                                                                     #
# The odds ratios for the alternative-specific constants              #
# bus and train indicate the relative likelihood of selecting         #
# these alternatives versus travelling by car (the omitted            #
# category), assuming that cost and time are the same for all         #
# modes.                                                              #
#                                                                     #
# For example, if cost and time were equal, individuals would be 4.36 #
# times more likely to travel by bus than by car, and they            #
# would be 14.46 times more likely to travel by train than by car.    #

choice.clogit.predictions    <- choice.clogit$fitted.values
table                        <- cbind(travel$train             ,
                                      travel$bus               ,
                                      travel$time              ,
                                      travel$invc              ,
                                      travel$choice            ,
                                      choice.clogit.predictions)
colnames(table) <- c("train" ,
                     "bus"   ,
                     "time"  ,
                     "invc"  ,
```

```
                      "choice",
                      "prob"  )

# First individual
table[1:3,]
```

```
     train bus time invc choice       prob
[1,]     1   0  406   31      0 0.06424769
[2,]     0   1  452   25      0 0.01072045
[3,]     0   0  180   10      1 0.92503186
```

```
# The predicted probability of travelling by car     #
# (the option chosen) is .93, while the predicted   #
# probability of travelling by train is only .06.    #
# In this case, the choice corresponds to choosing  #
# the cheapest and quickest mode of transportation. #
# If we consider another observation where          #
# train was chosen.                                 #

table[16:18,]
```

```
     train bus time invc choice       prob
[1,]     1   0  385   20      1 0.54937707
[2,]     0   1  452   13      0 0.06434809
[3,]     0   0  284   12      0 0.38627483
```

```
# we see that the probability of choosing train was estimated      #
# to be .55, while the probability of driving was .39.             #
# In this case, the respondent chose to                            #
# travel by train even though it was neither cheapest nor fastest. #
# WTP for time
# The ratio of the coefficients for time and invc is  #
# the marginal rate of substitution between time and  #
# cost.                                               #
cat(
  "WTP for time: ",
choice.clogit$coefficients[3]/choice.clogit$coefficients[4],"$/min", "\n")
```

```
WTP for time:  0.3974874 $/min
```

The quantity computed above is the **willingness to pay (WTP) for travel time savings**. In the conditional logit model, it is obtained as the ratio of the coefficient on travel time to the coefficient on travel cost:

$$WTP_{\text{time saving}} = \frac{\beta_{\text{time}}}{\beta_{\text{cost}}}.$$

Because both β_{time} and β_{cost} are negative, this ratio is positive. The interpretation is straightforward: it gives the amount of money an individual is willing to pay to reduce travel time by one minute, holding utility constant.

For example, if the estimated value is about 0.40, this means that individuals are willing to pay approximately \$0.40 for a one-minute reduction in travel time. Equivalently, this is about \$24 per hour of travel time savings.

This result summarizes the trade-off between time and money in the model. A larger WTP means that travelers place a higher value on saving time, whereas a smaller WTP means that cost considerations are relatively more important than time savings.

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